

APPENDIX A MATHEMATICAL DERIVATIONS AND WORKED EXAMPLE

This appendix consolidates all formal derivations, iterative procedures, and supporting mathematics referenced in the main body of the VCH Sizing Framework. Sections -A and -B establish foundational procedures used across multiple steps of the design framework. Sections -C through -J follow the step numbering of the framework itself. All equation numbers within this appendix are self-contained and cross-referenced from the main body by the labels shown in each section heading.

A. Linear Interpolation for Thermophysical Property Lookup

Thermophysical properties — density ρ , dynamic viscosity μ , specific heat capacity c_p , thermal conductivity k_f , and volumetric thermal expansion coefficient β — are evaluated at local fluid temperatures by linear interpolation over a discrete property table. This procedure is referenced in Steps 2, 5, 6, and 9 of the main framework.

Procedure: Let $\{T_j, \phi_j\}_{j=1}^M$ denote a tabulated property pair where T_j are monotonically increasing temperature nodes and ϕ_j are the corresponding property values. For a query temperature T^* satisfying $T_j \leq T^* \leq T_{j+1}$, the interpolated property is:

$$\phi(T^*) = \phi_j + \frac{T^* - T_j}{T_{j+1} - T_j} (\phi_{j+1} - \phi_j). \quad (1)$$

The bounding index j is located by a bisection search on the sorted temperature array in $O(\log M)$ comparisons. Extrapolation beyond the table range ($T^* < T_1$ or $T^* > T_M$) must be flagged as an error; the calculation should be aborted and the user required to supply data at the relevant temperature.

Validity and Accuracy Bounds: Equation (1) is valid provided:

- 1) T^* lies within the tabulated range: $T_1 \leq T^* \leq T_M$.
- 2) The property ϕ varies smoothly and without phase change across the interval $[T_j, T_{j+1}]$.
- 3) The node spacing $\Delta T_j = T_{j+1} - T_j$ is sufficiently small that the linear approximation introduces negligible error relative to the physical property variation.

For water and common heat-transfer fluids over industrial operating ranges, linear interpolation between nodes spaced at 10 K or less introduces errors below 0.5% for ρ , c_p , and k_f , and below 2% for μ away from the saturation boundary [1].

Dynamic viscosity μ exhibits the strongest nonlinearity with temperature, particularly for viscous fluids at low temperatures. Where high accuracy is required for μ , a log-linear interpolation of the form $\ln[\mu(T)]$ is recommended in place of (1).

B. Worked Numerical Example

The following end-to-end calculation traces a single representative industrial scenario through all eleven steps of the design framework. All intermediate values are stated explicitly so that a practising engineer may use this section as a verification reference when implementing the framework for the first time. All fluid properties are obtained via (1) applied to the tables in Appendix B.

Design Scenario:

- **Working fluid:** Water at coil inlet $T_{in} = 80^\circ\text{C}$, mass flow rate $\dot{m} = 0.5 \text{ kg s}^{-1}$.
- **Tank fluid:** Water at bulk temperature $T_\infty = 20^\circ\text{C}$.
- **Design duty:** $Q_{\text{design}} = 25 \text{ kW}$.
- **Vessel aspect ratio:** $\lambda = 1.5$, hemispherical heads.
- **Tube geometry:** $d_i = 25 \text{ mm}$, $d_o = 28 \text{ mm}$, $k_w = 16 \text{ W m}^{-1} \text{ K}^{-1}$ (austenitic stainless steel).
- **Coil centreline diameter:** $D_c = 0.30 \text{ m}$.
- **Insulation:** $t_{ins} = 50 \text{ mm}$, $k_{ins} = 0.04 \text{ W m}^{-1} \text{ K}^{-1}$ (mineral wool).

Step 1 — Fluid Properties (via Section -A): At film temperature $T_f = (80 + 20)/2 = 50^\circ\text{C}$: $\rho = 988 \text{ kg m}^{-3}$, $\mu = 5.47 \times 10^{-4} \text{ Pa s}$, $k_f = 0.644 \text{ W m}^{-1} \text{ K}^{-1}$, $c_p = 4181 \text{ J kg}^{-1} \text{ K}^{-1}$, $Pr = 3.55$.

Step 2 — Reynolds Number (Section -E1):

$$Re = \frac{4\dot{m}}{\pi d_i \mu} = \frac{4 \times 0.5}{\pi \times 0.025 \times 5.47 \times 10^{-4}} = 46,400 \quad (\text{turbulent}).$$

Step 3 — Inner Nusselt Number (Section -F2): Petukhov friction factor: $f = (0.790 \ln 46400 - 1.64)^{-2} = 0.02093$.

$$Nu_{\text{straight}} = \frac{(0.02093/8)(46400 - 1000) \times 3.55}{1 + 12.7\sqrt{0.02093/8}(3.55^{2/3} - 1)} = 183.4.$$

Curvature ratio $\delta = 0.028/0.30 = 0.0933$. Correction: $C_{\text{curv}} = 1 + 3.6(1 - 0.0933)(0.0933)^{0.8} = 1.252$.

$$Nu_{\text{coil}} = 183.4 \times 1.252 = 229.6, \quad h_i = \frac{229.6 \times 0.644}{0.025} = 5916 \text{ W m}^{-2} \text{ K}^{-1}.$$

Step 4 — Outer Natural Convection (Section -F5): At $\Delta T = 60$ K and film properties of water at 50°C : $Ra_{d_o} = g\beta\Delta T d_o^3/(\nu\alpha) = 4.27 \times 10^4$.

$$Nu_{\text{outer}} = \left[0.60 + \frac{0.387 \times (4.27 \times 10^4)^{1/6}}{[1 + (0.559/3.55)^{9/16}]^{8/27}} \right]^2 = 9.87, \quad h_o = \frac{9.87 \times 0.644}{0.028} = 226.8 \text{ W m}^{-2} \text{ K}^{-1}.$$

Step 5 — Overall U (Section -F1):

$$\begin{aligned} \frac{1}{U} &= \frac{r_o}{h_i r_i} + \frac{r_o \ln(r_o/r_i)}{k_w} + \frac{1}{h_o} \\ &= \frac{0.014}{5916 \times 0.0125} + \frac{0.014 \ln(1.120)}{16} + \frac{1}{226.8} \\ &= 1.890 \times 10^{-4} + 9.83 \times 10^{-5} + 4.409 \times 10^{-3} = 4.726 \times 10^{-3} \text{ m}^2 \text{ K W}^{-1}, \\ U &= 211.6 \text{ W m}^{-2} \text{ K}^{-1}. \end{aligned}$$

Step 6 — NTU_{req} (Section -G1): $C_{\min} = 0.5 \times 4181 = 2090.5 \text{ W K}^{-1}$.

$$\varepsilon_{\text{target}} = \frac{25000}{2090.5 \times 60} = 0.1994, \quad NTU_{\text{req}} = -\ln(1 - 0.1994) = 0.2228.$$

Step 7 — Required Area and Coil Length (Sections -G2 and -G3):

$$A_{o,\text{req}} = \frac{0.2228 \times 2090.5}{211.6} = 2.202 \text{ m}^2, \quad L_{\text{coil}} = \frac{2.202}{\pi \times 0.028} = 25.02 \text{ m}.$$

Step 8 — Coil Pitch and Turn Count (Sections -G4 and -G5): With $h_1 = 0.80$ m and $D_c = 0.30$ m, the closed-form pitch from (71):

$$p_{\text{coil}} = \frac{\pi \times 0.30 \times 0.80}{\sqrt{25.02^2 - 0.80^2}} = \frac{0.7540}{25.01} = 30.1 \text{ mm}.$$

Turn arc length: $\ell = \sqrt{(\pi \times 0.30)^2 + 0.0301^2} = 0.9426$ m. $N_t = [25.02/0.9426] = 27$ turns. Winding height: $h_{\text{wind}} = 27 \times 0.0301 = 0.813$ m.

Since $h_{\text{wind}} > h_1 = 0.80$ m, D_c must be increased marginally and the Newton–Raphson procedure of Algorithm -G4 applied. This confirms the iterative procedure is required in general.

Step 9 — Displacement Compensation (Section -H1):

$$V_{\text{disp}} = \frac{\pi \times 0.028^2}{4} \times 25.02 = 0.01541 \text{ m}^3 = 15.4 \text{ L}.$$

Step 10 — Steady-State Heat Loss (Section -I5): With $t_w = 3$ mm, $k_s = 16 \text{ W m}^{-1} \text{ K}^{-1}$, $D_i = 0.50$ m, $L_s = 0.80$ m:

$$R_{\text{wall}} = \frac{\ln(1.012)}{2\pi \times 16 \times 0.80} = 1.47 \times 10^{-4} \text{ K W}^{-1}, \quad R_{\text{ins}} = \frac{\ln(1.20)}{2\pi \times 0.04 \times 0.80} = 0.7272 \text{ K W}^{-1}.$$

Neglecting inner and outer convective resistances (small relative to insulation): $R_{\text{tot}} \approx 0.7274 \text{ K W}^{-1}$, $\dot{Q}_{\text{loss}} = (80 - 20)/0.7274 = 82.5 \text{ W}$.

Step 11 — Thermal Battery Sizing (Section -J5): $\dot{Q}_{\text{total}} = 82.5 + 25000 = 25083 \text{ W}$, $t_{\text{batch}} = 3600$ s, $\Delta T_{\text{store}} = T_{\text{max}} - T_{\text{min}} = 10$ K:

$$V_{\text{max}} = \frac{25083 \times 3600}{988 \times 4181 \times 10} = 2.189 \text{ m}^3 = 2189 \text{ L}.$$

Thermal time constant: $\tau_{\text{th}} = 988 \times 2.189 \times 4181 \times 0.7274 = 6.574 \times 10^6 \text{ s} \approx 76$ days, confirming excellent thermal autonomy for this vessel size.

C. Vessel Geometry Derivations (Step 3)

1) Algebraic Solution for Vessel Dimensions — Conical-Capped Vessel:

Geometry Definition: The vessel comprises a vertical cylinder of internal diameter d and cylindrical wall height h_2 , capped at the base by a right circular cone of half-angle θ . The aspect ratio is defined as:

$$\alpha = \frac{h_2}{d}, \quad \alpha > 0. \quad (2)$$

The conical cap height follows from the half-angle:

$$h_c = \frac{d}{2} \tan \theta. \quad (3)$$

TABLE I
WORKED EXAMPLE — SUMMARY OF KEY RESULTS

Step	Quantity	Symbol	Value
2	Reynolds number	Re	46 400
3	Inner heat transfer coeff.	h_i	$5916 \text{ W m}^{-2} \text{ K}^{-1}$
4	Outer heat transfer coeff.	h_o	$226.8 \text{ W m}^{-2} \text{ K}^{-1}$
5	Overall heat transfer coeff.	U	$211.6 \text{ W m}^{-2} \text{ K}^{-1}$
6	Required NTU	NTU_{req}	0.2228
7	Required coil length	L_{coil}	25.02 m
8	Helix pitch	p_{coil}	30.1 mm
9	Coil displacement volume	V_{disp}	15.4 L
10	Steady-state heat loss	$\dot{Q}_{\text{loss}}$	82.5 W
11	Thermal battery max. volume	V_{max}	2189 L

Volume Formula and Cubic Solution: The total internal volume is the sum of the cylindrical and conical contributions:

$$V = \frac{\pi d^2}{4} h_2 + \frac{\pi d^2}{12} h_c. \quad (4)$$

Substituting (2) and (3) into (4) and collecting terms:

$$V = \frac{\pi d^3}{24} (6\alpha + \tan \theta). \quad (5)$$

Setting $V = V_{\text{target}}$ and solving for d :

$$d = \left(\frac{24 V_{\text{target}}}{\pi (6\alpha + \tan \theta)} \right)^{1/3}. \quad (6)$$

The cylindrical wall height and conical cap height follow immediately:

$$h_2 = \alpha d, \quad (7)$$

$$h_c = \frac{d}{2} \tan \theta. \quad (8)$$

Equation (6) is a closed-form cube root — no iterative solution is required for the conical-capped geometry. This distinguishes it from the hemi-ellipsoidal case in Section -C2, where the cap volume coefficient differs.

2) *Algebraic Solution for Vessel Dimensions — Hemi-Ellipsoidal-Capped Vessel:*

Geometry Definition: The base cap is a hemi-ellipsoid with equatorial radius $d/2$ and polar semi-axis b . The ratio $\beta = 2b/d$ defines the cap slenderness. A standard shallow (2:1 ellipsoidal) cap uses $\beta = 0.25$; a hemisphere corresponds to $\beta = 1$.

Volume Formula and Cubic Solution: The volume of a hemi-ellipsoid of equatorial radius $d/2$ and polar semi-axis $b = \beta d/2$ is:

$$V_{\text{cap}} = \frac{\pi d^3 \beta}{12}. \quad (9)$$

Total vessel volume:

$$V = \frac{\pi d^2}{4} (\alpha d) + \frac{\pi d^3 \beta}{12} = \frac{\pi d^3}{12} (3\alpha + \beta). \quad (10)$$

Explicit solution for d :

$$d = \left(\frac{12 V_{\text{target}}}{\pi (3\alpha + \beta)} \right)^{1/3}, \quad (11)$$

with $h_2 = \alpha d$ and cap polar semi-axis $b = \beta d/2$.

3) *Conical Cap Lateral Surface Area from First Principles:* A right circular cone of base radius $r = d/2$ and half-angle θ has slant height:

$$\ell_s = \frac{r}{\sin \theta} = \frac{d}{2 \sin \theta}. \quad (12)$$

The lateral surface area is obtained by unrolling the cone into a flat sector of radius ℓ_s and arc length $2\pi r$:

$$A_{\text{cone}} = \pi r \ell_s = \frac{\pi d^2}{4 \sin \theta}. \quad (13)$$

Equation (13) excludes the base circle area πr^2 , which is not an external surface in a closed vessel. The cylindrical wall lateral area is $A_{\text{cyl}} = \pi d h_2$.

4) *Hemi-Ellipsoidal Cap Surface Area and Working Approximation:*

Exact Formula via Elliptic Integral: For a hemi-ellipsoid with equatorial radius $a = d/2$ and polar semi-axis b , defining eccentricity $e = \sqrt{1 - (b/a)^2}$ for the oblate case ($b < a$):

$$A_{\text{hemi,exact}} = 2\pi a^2 \left(1 + \frac{1 - e^2}{e} \operatorname{arctanh}(e) \right). \quad (14)$$

For the prolate case ($b > a$), with $e = \sqrt{1 - (a/b)^2}$:

$$A_{\text{hemi,prolate}} = 2\pi a^2 \left(1 + \frac{b}{ae} \arcsin(e) \right). \quad (15)$$

Knud Thomsen Working Approximation: For engineering calculations the Knud Thomsen approximation is used with $p = 1.6075$ [12]:

$$A_{\text{hemi,approx}} \approx 2\pi \left(\frac{2(d/2)^p b^p + (d/2)^{2p}}{3} \right)^{1/p}. \quad (16)$$

This approximation has a maximum relative error of 1.061% across all ellipsoid axis ratios [12], well within the fabrication tolerance of industrial pressure vessels (typically $\pm 1.5\%$ on wall thickness).

D. NPS Selection and Maximum Coil Configuration (Step 4)

1) *Target Volumetric Flow Rate from Thermal Demand:* The heating fluid must deliver thermal power $\dot{Q}_{\text{design}}$ to the process. For a single-phase fluid with no phase change, the sensible heat equation is:

$$\dot{Q}_{\text{design}} = \dot{m}_{\text{hot}} c_{p,\text{hot}} \Delta T_{\text{coil}}, \quad (17)$$

where $\Delta T_{\text{coil}} = T_{\text{hot,in}} - T_{\text{hot,out}}$ is the hot-side temperature drop. Solving for mass flow rate and converting to volumetric flow rate using density ρ_{hot} at the mean coil temperature:

$$\dot{V} = \frac{\dot{m}_{\text{hot}}}{\rho_{\text{hot}}} = \frac{\dot{Q}_{\text{design}}}{\rho_{\text{hot}} c_{p,\text{hot}} \Delta T_{\text{coil}}}. \quad (18)$$

All fluid properties are evaluated at $T_{\text{mean}} = (T_{\text{hot,in}} + T_{\text{hot,out}})/2$ using (1).

2) *Minimum Internal Flow Area from Velocity Ceiling Constraint:* The continuity equation relates volumetric flow rate to mean velocity u :

$$\dot{V} = u \frac{\pi d_i^2}{4}. \quad (19)$$

Enforcing $u \leq u_{\text{max}}$, the minimum required internal flow area is:

$$A_{\text{flow,min}} = \frac{\dot{V}}{u_{\text{max}}}, \quad (20)$$

and the corresponding minimum internal diameter:

$$d_{i,\text{min}} = \sqrt{\frac{4 \dot{V}}{\pi u_{\text{max}}}}. \quad (21)$$

The NPS schedule is selected as the smallest standard size with $d_i \geq d_{i,\text{min}}$. The actual mean velocity is back-calculated as:

$$u = \frac{4 \dot{V}}{\pi d_i^2} \leq u_{\text{max}}. \quad (22)$$

3) *Helical Coil Arc Length from Parametric Integration:*

Parametric Definition: A helical coil of N turns, centreline diameter D_c , and axial pitch p is described by:

$$x(\varphi) = \frac{D_c}{2} \cos \varphi, \quad (23)$$

$$y(\varphi) = \frac{D_c}{2} \sin \varphi, \quad (24)$$

$$z(\varphi) = \frac{p}{2\pi} \varphi, \quad (25)$$

where $\varphi \in [0, 2\pi N]$ is the azimuthal parameter.

Arc Length Integration: The arc length element is:

$$ds = \sqrt{\left(\frac{dx}{d\varphi}\right)^2 + \left(\frac{dy}{d\varphi}\right)^2 + \left(\frac{dz}{d\varphi}\right)^2} d\varphi. \quad (26)$$

Substituting the derivatives of (23)–(25) and applying $\sin^2 \varphi + \cos^2 \varphi = 1$:

$$ds = \sqrt{\frac{D_c^2}{4} + \frac{p^2}{4\pi^2}} d\varphi. \quad (27)$$

Since the integrand is constant in φ , integrating over N complete turns:

$$L_{\text{coil}} = N \sqrt{(\pi D_c)^2 + p^2}. \quad (28)$$

Equation (28) uses the coil *centreline* diameter $D_c = d_{\text{tank}} - 0.200 - d_o$ where the 0.200 m term accounts for the mandatory 100 mm radial clearance on each side. All dimensions must be in consistent units of metres.

4) *Compound Packing Efficiency and Practical Coil Length:* The geometric maximum coil length from (28) assumes ideal continuous winding with no manufacturing or operational constraints. The practical maximum is obtained by applying the compound packing efficiency factor η_{pack} , which is the product of five independent sub-factors, each addressing a distinct physical constraint:

$$\eta_{\text{pack}} = \eta_f \cdot \eta_s \cdot \eta_w \cdot \eta_t \cdot \eta_d. \quad (29)$$

Reference values for each sub-factor are tabulated in the main framework (Step 4, Tables 1 through 5). Typical compound values across standard industrial coil designs fall in the range $\eta_{\text{pack}} \in [0.55, 0.75]$.

The practical maximum coil length is:

$$L_{\text{practical}} = L_{\text{max}} \times \eta_{\text{pack}}, \quad (30)$$

where L_{max} is evaluated from (28) at the minimum structural pitch $p_{\text{min}} = 2d_o$ and maximum number of turns $N_{\text{max}} = \lfloor h_1/p_{\text{min}} \rfloor$.

E. Hydraulic Resistance and Pump Selection (Step 5)

1) *Reynolds Number for Internal Pipe Flow:*

$$Re = \frac{\rho u d_i}{\mu} = \frac{4 \dot{m}}{\pi d_i \mu}. \quad (31)$$

All fluid properties are evaluated at bulk mean temperature via (1). Flow regime classification for straight pipe: $Re < 2300$ laminar; $2300 \leq Re \leq 4000$ transitional; $Re > 4000$ turbulent. These boundaries are modified for helical geometry by the Dean number (Section -E3).

2) *Colebrook-White Equation and Newton-Raphson Solution:*

Colebrook-White Equation: For turbulent flow in a pipe of internal roughness ε :

$$\frac{1}{\sqrt{f_s}} = -2 \log_{10} \left(\frac{\varepsilon/d_i}{3.7} + \frac{2.51}{Re \sqrt{f_s}} \right), \quad (32)$$

where f_s denotes the straight-pipe Darcy friction factor. For laminar flow ($Re < 2300$):

$$f_s = \frac{64}{Re}. \quad (33)$$

Haaland Seed Estimate:

$$f_0 = \left(-1.8 \log_{10} \left[\left(\frac{\varepsilon/d_i}{3.7} \right)^{1.11} + \frac{6.9}{Re} \right] \right)^{-2}. \quad (34)$$

Newton-Raphson Iterative Procedure: Define $x = f_s^{-1/2}$ and the residual function $F(x) = 0$:

$$F(x) = x + 2 \log_{10} \left(\frac{\varepsilon/d_i}{3.7} + \frac{2.51 x}{Re} \right), \quad (35)$$

with derivative:

$$F'(x) = 1 + \frac{2}{\ln 10} \cdot \frac{2.51/Re}{\frac{\varepsilon/d_i}{3.7} + \frac{2.51 x}{Re}}. \quad (36)$$

The Newton-Raphson update at iteration k :

$$x^{(k+1)} = x^{(k)} - \frac{F(x^{(k)})}{F'(x^{(k)})}. \quad (37)$$

Initialise with $x^{(0)} = f_0^{-1/2}$ from (34). Terminate when $|x^{(k+1)} - x^{(k)}| < 10^{-8}$. Recover friction factor: $f_s = 1/(x^{(k+1)})^2$. Convergence is achieved in three or fewer iterations across the full turbulent regime [7].

3) *Dean Number Definition and Flow Regime Classification:* The Dean number quantifies the ratio of centrifugal to viscous forces in helical pipe flow [8]:

$$De = Re \sqrt{\frac{d_i}{D_c}}. \quad (38)$$

Curvature stabilises the boundary layer, delaying the laminar-to-turbulent transition. The critical Dean number below which flow remains laminar despite $Re > 2300$ is $De_{crit} = 11.6$ [9], corresponding to:

$$Re_{crit,coil} = 11.6 \sqrt{\frac{D_c}{d_i}}. \quad (39)$$

Flow regime classification for helical coils:

- $De < 11.6$: laminar — use Sieder-Tate (Section -F3).
- $11.6 \leq De \leq 100$: transitional — interpolate between laminar and turbulent Ito correlations.
- $De > 100$: turbulent — use Gnielinski modified (Section -F2).

4) *Ito's Helical Friction Factor Correlations:*

Laminar Regime ($De < 11.6$):

$$f_{hel,lam} = \frac{64}{Re} \left[1 + 0.033 (\log_{10} De)^4 \right]. \quad (40)$$

Turbulent Regime ($De > 100$):

$$f_{hel,turb} = 0.304 Re^{-0.25} \left(\frac{D_c}{d_i} \right)^{-0.1}. \quad (41)$$

Transitional Regime ($11.6 \leq De \leq 100$): Linear interpolation in De :

$$f_{hel,trans} = f_{hel,lam}|_{De=11.6} + \frac{De - 11.6}{88.4} (f_{hel,turb}|_{De=100} - f_{hel,lam}|_{De=11.6}). \quad (42)$$

5) *Darcy-Weisbach Major Loss Applied to Helical Coil Geometry:*

$$\Delta P_{major} = f_{hel} \frac{L_{coil}}{d_i} \frac{\rho u^2}{2}, \quad (43)$$

where f_{hel} is the appropriate Ito friction factor from Section -E4 and L_{coil} is from (28).

6) *Minor Losses, Total System Resistance, and Head Conversion:*

K-Factor Method for Minor Losses: Each fitting contributes:

$$\Delta P_{minor,j} = K_j \frac{\rho u^2}{2}. \quad (44)$$

Total system pressure drop:

$$\Delta P_{system} = \left(f_{hel} \frac{L_{coil}}{d_i} + \sum_{j=1}^M K_j \right) \frac{\rho u^2}{2}. \quad (45)$$

Conversion to Equivalent Head:

$$H_{system} = \frac{\Delta P_{system}}{\rho g}, \quad (46)$$

where $g = 9.81 \text{ m s}^{-2}$.

7) *Pump Affinity Laws and VFD Frequency Scaling:* For a geometrically similar pump operating at rotational speed n , the affinity laws follow from invariance of the dimensionless flow, head, and power coefficients [3]:

$$\frac{\dot{V}_2}{\dot{V}_1} = \frac{n_2}{n_1}, \quad (47)$$

$$\frac{H_2}{H_1} = \left(\frac{n_2}{n_1} \right)^2, \quad (48)$$

$$\frac{P_2}{P_1} = \left(\frac{n_2}{n_1} \right)^3. \quad (49)$$

A Variable Frequency Drive (VFD) scales pump speed proportionally to supply frequency f_{VFD} . The target VFD frequency required to achieve velocity-compliant flow rate $\dot{V}_{design}$ is:

$$f_{VFD} = f_{rated} \frac{\dot{V}_{design}}{\dot{V}_{rated}}. \quad (50)$$

The scaled duty head is verified as:

$$H_{\text{duty}} = H_{\text{rated}} \left(\frac{f_{\text{VFD}}}{f_{\text{rated}}} \right)^2 \geq H_{\text{system}}. \quad (51)$$

If (51) is not satisfied, a larger NPS or pump frame is selected and the procedure repeats from Section -D2.

F. Thermal Performance Evaluation (Step 6)

1) *Series Thermal Resistance Network and Overall Heat Transfer Coefficient:* For a helical coil of inner radius r_i , outer radius r_o , wall conductivity k_w , and length L , the overall heat transfer coefficient referred to the outer surface is obtained from the series resistance network (inner convection, cylindrical wall conduction, outer convection):

$$\frac{1}{U} = \frac{r_o}{h_i r_i} + \frac{r_o \ln(r_o/r_i)}{k_w} + \frac{1}{h_o}. \quad (52)$$

Including fouling resistances $R_{f,i}$ and $R_{f,o}$ on the inner and outer surfaces respectively:

$$\frac{1}{U} = \frac{r_o}{h_i r_i} + R_{f,i} \frac{r_o}{r_i} + \frac{r_o \ln(r_o/r_i)}{k_w} + R_{f,o} + \frac{1}{h_o}. \quad (53)$$

2) *Gnielinski Correlation Modified for Helical Coil Turbulent Flow:* The Gnielinski correlation for a straight tube ($3000 \lesssim Re \lesssim 5 \times 10^6$, $0.5 \lesssim Pr \lesssim 2000$) is:

$$Nu_{\text{straight}} = \frac{(f/8)(Re - 1000)Pr}{1 + 12.7\sqrt{f/8}(Pr^{2/3} - 1)}, \quad (54)$$

where f is from the Petukhov relation $f = (0.790 \ln Re - 1.64)^{-2}$.

For a helical coil with curvature ratio $\delta = d_i/D_c$, the Manlapaz-Churchill curvature correction [10] gives:

$$Nu_{\text{coil}} = Nu_{\text{straight}} \left[1 + 3.6 \left(1 - \frac{d_i}{D_c} \right) \left(\frac{d_i}{D_c} \right)^{0.8} \right], \quad (55)$$

and $h_i = Nu_{\text{coil}} k_f / d_i$.

3) *Sieder-Tate Correlation for Laminar and Transitional Helical Coil Flow:* For laminar and transitional flow, substituting the Dean number for Re to account for secondary-flow enhancement [10]:

$$Nu_{\text{coil}} = 1.86 \left(\frac{De Pr d_i}{L_{\text{coil}}} \right)^{1/3} \left(\frac{\mu_b}{\mu_w} \right)^{0.14}, \quad (56)$$

valid for Graetz number $Gz = Re Pr d_i / L_{\text{coil}} > 10$ and $De < 11.6$. Here μ_b and μ_w are the dynamic viscosities at bulk and wall temperatures respectively.

4) *Prandtl Number Definition and Role in Convective Correlations:*

$$Pr = \frac{\mu c_p}{k_f} = \frac{\nu}{\alpha_{\text{th}}}, \quad (57)$$

where $\nu = \mu/\rho$ is kinematic viscosity and $\alpha_{\text{th}} = k_f/(\rho c_p)$ is thermal diffusivity. The Prandtl number governs the relative thickness of the hydrodynamic and thermal boundary layers: $\delta_{\text{th}}/\delta_{\text{hydro}} \approx Pr^{-1/3}$.

5) *Churchill-Chu Natural Convection Correlation for a Horizontal Cylinder:* Treating each coil turn as a long horizontal cylinder of outer diameter d_o submerged in the bulk tank fluid [11]:

$$\overline{Nu}_{d_o} = \left\{ 0.60 + \frac{0.387 Ra_{d_o}^{1/6}}{[1 + (0.559/Pr)^{9/16}]^{8/27}} \right\}^2, \quad (58)$$

valid for $Ra_{d_o} \lesssim 10^{12}$. The Rayleigh number, with all fluid properties evaluated at the film temperature $T_f = (T_{\text{wall}} + T_{\infty})/2$:

$$Ra_{d_o} = \frac{g \beta \Delta T d_o^3}{\nu \alpha_{\text{th}}}, \quad (59)$$

and $h_o = \overline{Nu}_{d_o} k_f / d_o$.

6) *Logarithmic Mean Temperature Difference — Derivation and Arithmetic Mean Approximation:*

Derivation: For a coil immersed in a well-mixed tank at constant temperature T_{∞} , the energy balance on a differential element $dA = \pi d_o dx$ gives $d\theta/\theta = -(U \pi d_o / \dot{m} c_p) dx$, where $\theta = T_f - T_{\infty}$. Integrating over the full coil length:

$$q = U A_o \Delta T_{\text{lm}}, \quad \Delta T_{\text{lm}} = \frac{\Delta T_1 - \Delta T_2}{\ln(\Delta T_1 / \Delta T_2)}, \quad (60)$$

where $\Delta T_1 = T_{\text{in}} - T_{\infty}$ and $\Delta T_2 = T_{\text{out}} - T_{\infty}$.

Arithmetic Mean Approximation Validity: The arithmetic mean $\Delta T_{\text{am}} = (\Delta T_1 + \Delta T_2)/2$ approximates ΔT_{lm} within 5% when the ratio $R = \Delta T_1 / \Delta T_2$ satisfies $0.54 \lesssim R \lesssim 1.86$, i.e. when the two endpoint temperature differences differ by less than a factor of approximately 1.86.

7) *Effectiveness-NTU Method: Full Derivation for $C_r = 0$:* For a submerged coil in a large well-mixed tank, the tank thermal capacity rate $C_{\text{tank}} \rightarrow \infty$, giving capacity ratio $C_r = C_{\text{min}}/C_{\text{max}} = 0$. The energy balance on the coil stream reduces to the ODE $d\theta/dA = -(U/C_{\text{min}})\theta$, where $\theta = T_{\text{min}} - T_{\infty}$. Integrating:

$$\ln\left(\frac{\theta_{\text{out}}}{\theta_{\text{in}}}\right) = -NTU, \quad NTU = \frac{UA_o}{C_{\text{min}}}. \quad (61)$$

The heat exchanger effectiveness:

$$\varepsilon = \frac{q}{q_{\text{max}}} = \frac{C_{\text{min}}(T_{\text{min,out}} - T_{\text{min,in}})}{C_{\text{min}}(T_{\infty} - T_{\text{min,in}})}, \quad (62)$$

simplifies, after substituting the integrated ODE result, to:

$$\boxed{\varepsilon = 1 - e^{-NTU}, \quad (C_r = 0).} \quad (63)$$

8) *Maximum Transmissible Thermal Power:*

$$q_{\text{max}} = C_{\text{min}} (T_{h,\text{in}} - T_{\infty}) = \dot{m} c_p (T_{\text{in}} - T_{\infty}). \quad (64)$$

Combining with (63):

$$\boxed{q = (1 - e^{-UA_o/C_{\text{min}}}) C_{\text{min}} (T_{\text{in}} - T_{\infty}).} \quad (65)$$

This expression encapsulates coil geometry (through A_o), thermal resistances (through U from (52) or (53)), flow conditions (through C_{min}), and driving potential (through $T_{\text{in}} - T_{\infty}$). It is evaluated twice in Step 6: once using U_{steady} for nominal performance, and once using U_{cold} (worst-case cold-start viscosity) for conservative sizing.

G. Coil Geometry Optimisation (Step 7)

1) *Target Effectiveness and NTU Back-Calculation from Q_{design} :*

$$\varepsilon_{\text{target}} = \frac{Q_{\text{design}}}{q_{\text{max}}} = \frac{Q_{\text{design}}}{C_{\text{min}} (T_{\text{in}} - T_{\infty})}, \quad (66)$$

with $0 < \varepsilon_{\text{target}} < 1$ required for physical validity. From (63):

$$\boxed{NTU_{\text{req}} = -\ln(1 - \varepsilon_{\text{target}}).} \quad (67)$$

2) *Required Surface Area from NTU_{req} and U_{cold} :* From $NTU = UA_o/C_{\text{min}}$:

$$\boxed{A_{o,\text{req}} = \frac{NTU_{\text{req}} C_{\text{min}}}{U_{\text{cold}}}.} \quad (68)$$

U_{cold} is computed from (53) using fluid properties evaluated at the cold-start temperature (worst-case viscosity from Step 2).

3) *Optimised Coil Length from Required Surface Area:*

$$\boxed{L_{\text{coil}} = \frac{A_{o,\text{req}}}{\pi d_o}.} \quad (69)$$

This must satisfy $L_{\text{coil}} \leq L_{\text{practical}}$ from (30). If this constraint is violated, the NPS is incremented upward and Steps 4 through 7 are repeated.

4) *Helical Pitch Inversion — Transcendental Equation and Newton-Raphson Solution:*

Closed-Form Solution: Given total coil length L_{coil} , coil centreline diameter D_c , and winding height h_1 , the helical geometry relation (28) with $N = h_1/p$ gives:

$$\frac{h_1}{p} \sqrt{(\pi D_c)^2 + p^2} = L_{\text{coil}}. \quad (70)$$

Squaring and solving algebraically:

$$\boxed{p_{\text{coil}} = \frac{\pi D_c h_1}{\sqrt{L_{\text{coil}}^2 - h_1^2}}}, \quad (71)$$

valid when $L_{\text{coil}} > h_1$. When additional geometric constraints couple p to D_c , the full iterative procedure below is required.

Newton-Raphson Iterative Procedure: Define $F(p)$ as in (70). Its derivative is:

$$F'(p) = \frac{-h_1 (\pi D_c)^2}{p^2 \sqrt{(\pi D_c)^2 + p^2}}. \quad (72)$$

[H] $D_c, h_1, L_{\text{coil}}, d_o, \delta_{\text{min}}$ (min gap), $\tau = 10^{-6} \text{ m}$ $p_{\text{coil}} \leftarrow p \leftarrow d_o(1 + \delta_{\text{min}})$ *minimum admissible pitch $|\Delta p| < \tau$ $F \leftarrow (h_1/p) \sqrt{(\pi D_c)^2 + p^2} - L_{\text{coil}}$ $F' \leftarrow -h_1 (\pi D_c)^2 / [p^2 \sqrt{(\pi D_c)^2 + p^2}]$ $\Delta p \leftarrow -F/F'$ $p \leftarrow p + \Delta p$ $p < d_o(1 + \delta_{\text{min}})$ $p \leftarrow d_o(1 + \delta_{\text{min}})$ *enforce lower bound $p_{\text{coil}} \leftarrow p$ Newton-Raphson solver for helical coil pitch

Convergence is quadratic, typically requiring fewer than ten iterations. If $F'(p) \rightarrow 0$ near $p = \pi D_c$, switch to bisection on $[d_o(1 + \delta_{\text{min}}), h_1/2]$.

Pitch Sanity Bounds: The back-calculated pitch must satisfy:

$$p_{\min} = 2 d_o \leq p_{\text{coil}} \leq \frac{h_1}{3}. \quad (73)$$

If $p_{\text{coil}} < p_{\min}$, the coil is already at maximum density and cannot be reduced further without mechanical clashing (see Step 7, Section 7.4.1 of the main framework). If $p_{\text{coil}} > h_1/3$, fewer than three turns result and the helical flow assumption is no longer valid.

5) *Turn Count, Winding Height, and End Clearance Formulations:* Turn arc length: $\ell_{\text{turn}} = \sqrt{(\pi D_c)^2 + p_{\text{coil}}^2}$.

Number of turns (rounded down to ensure the coil fits within h_1):

$$N_t = \left\lfloor \frac{h_1}{p_{\text{coil}}} \right\rfloor. \quad (74)$$

Actual winding height:

$$h_{\text{wind}} = N_t p_{\text{coil}}. \quad (75)$$

End clearance at each extremity:

$$h_{\text{clearance}} = h_1 - h_{\text{wind}} \geq 50 \text{ mm}. \quad (76)$$

If (76) is violated, N_t is reduced by one and the thermal adequacy check of Step 7 is repeated with the reduced surface area.

H. Volumetric Displacement Compensation (Step 8)

1) *Primary Coil Pipe Displacement Volume:* The coil displaces volume equal to the full external cross-section of the tube, since the tube interior (thermofluid) is inaccessible to the process fluid. Using the *external* diameter d_o :

$$V_{\text{disp}} = \frac{\pi d_o^2}{4} L_{\text{coil}}. \quad (77)$$

Using d_i instead of d_o would underestimate the displacement by the tube wall volume and is incorrect.

2) *Gross Fabrication Volume Target:*

$$V_{\text{gross}} = V_{\text{target}} + V_{\text{disp}} + V_{\text{ancillary}}, \quad (78)$$

where $V_{\text{ancillary}} = k_{\text{anc}} \times V_{\text{disp}}$ with k_{anc} from the support complexity table in Step 8.

3) *Option A — Additional Shell Height for Vertical Extension:* With vessel diameter d held constant, the additional cylindrical height required to recover $V_{\text{gross}} - V_{\text{target}}$:

$$\Delta h = \frac{4(V_{\text{disp}} + V_{\text{ancillary}})}{\pi d^2}, \quad h_{1,\text{final}} = h_1 + \Delta h. \quad (79)$$

4) *Option B — Adjusted Vessel Diameter for Radial Expansion:* With shell height h_1 held constant, the adjusted diameter is obtained by substituting V_{gross} for V_{target} in the Step 3 cubic solution. For the conical-capped case from (6):

$$d_{\text{new}} = \left(\frac{24 V_{\text{gross}}}{\pi(6\alpha + \tan \theta)} \right)^{1/3}. \quad (80)$$

For the hemi-ellipsoidal case, substitute V_{gross} into (11).

5) *Updated External Surface Area After Dimensional Adjustment:*

Option A (diameter d fixed, new height $h_{1,\text{final}}$):

$$A_{\text{shell,A}} = \pi d h_{1,\text{final}}, \quad A_{\text{total,A}} = A_{\text{shell,A}} + A_{\text{caps}}. \quad (81)$$

A_{caps} is evaluated using (13) or (16) as appropriate and is unchanged from Step 3.

Option B (height h_1 fixed, new diameter d_{new}): Both shell and cap areas must be recomputed with d_{new} in place of d , using the same surface area formulas from Section -C3 or -C4.

I. Thermal Driver Sizing (Step 9)

1) *Internal Natural Convection Resistance at the Vessel Boundary:* The inner vessel wall is treated as a vertical flat plate or large-diameter horizontal cylinder. Applying the Churchill-Chu correlation (58) with characteristic length equal to the vessel inner diameter D_i and $\Delta T = T_{\text{bulk}} - T_{\text{wall,i}}$:

$$h_{\text{nc,i}} = \frac{Nu_{D_i} k_f}{D_i}, \quad (82)$$

and the corresponding resistance referred to inner surface area A_i :

$$R_{\text{conv,i}} = \frac{1}{h_{\text{nc,i}} A_i}. \quad (83)$$

For conservative preliminary estimation, $h_{\text{nc,i}}$ may be taken as $50\text{--}200 \text{ W m}^{-2} \text{ K}^{-1}$ for low-to-moderate viscosity process fluids and $10\text{--}50 \text{ W m}^{-2} \text{ K}^{-1}$ for high-viscosity organic fluids.

2) *Vessel Wall Conductive Resistance — Flat Wall and Cylindrical Formulations:*

Cylindrical Formulation:

$$R_{\text{wall,cyl}} = \frac{\ln(r_o/r_i)}{2\pi k_s L_s}. \quad (84)$$

Flat Wall Approximation:

$$R_{\text{wall,flat}} = \frac{t_w}{k_s A_{\text{wall}}}. \quad (85)$$

The flat-wall approximation introduces less than 5% error when $t_w/r_i < 0.1$ (thin-walled condition).

3) *Insulation Conductive Resistance — Flat Wall and Cylindrical Formulations:*

Cylindrical Formulation:

$$R_{\text{ins,cyl}} = \frac{\ln(r_{\text{ins}}/r_o)}{2\pi k_{\text{ins}} L_s}, \quad (86)$$

where $r_{\text{ins}} = r_o + t_{\text{ins}}$.

Flat Wall Approximation:

$$R_{\text{ins,flat}} = \frac{t_{\text{ins}}}{k_{\text{ins}} A_{\text{ext}}}, \quad (87)$$

where A_{ext} is from Section -H5. The cylindrical form should be preferred when insulation thickness is not negligible relative to the vessel radius.

4) *External Convective Resistance Formulation:*

$$R_{\text{conv,o}} = \frac{1}{h_{\text{ext}} A_{\text{ins}}}, \quad (88)$$

where A_{ins} is the outer insulation surface area and h_{ext} is estimated from the Churchill-Chu correlation applied to the outer insulation geometry, or taken from the reference values in Step 9, Section 9.2.4 of the main framework.

5) *Total Series Resistance Network and Steady-State Heat Loss Rate:*

$$R_{\text{tot}} = R_{\text{conv,i}} + R_{\text{wall}} + R_{\text{ins}} + R_{\text{conv,o}}. \quad (89)$$

$$\dot{Q}_{\text{loss}} = \frac{T_{\text{bulk}} - T_{\text{worst,amb}}}{R_{\text{tot}}}. \quad (90)$$

$T_{\text{worst,amb}}$ is the worst-case ambient temperature established in Step 1 — coldest recorded for heating applications, hottest for cooling.

6) *Multi-Vessel Total Thermal Driver Capacity:* For N_v vessels, each with heat loss $\dot{Q}_{\text{loss},j}$ and process heat demand $Q_{\text{design},j}$:

$$\dot{Q}_{\text{driver,total}} = \sum_{j=1}^{N_v} (\dot{Q}_{\text{loss},j} + Q_{\text{design},j}). \quad (91)$$

A diversity factor $f_d \leq 1$ may be applied if simultaneous peak demand across all vessels is not required by the operating schedule.

7) *Driver Turndown Ratio and Heating Element Count Formulations:* Minimum number of elements to cover peak demand at rated output per element $\dot{Q}_{\text{rated}}$:

$$N_{e,\text{max}} = \left\lceil \frac{\dot{Q}_{\text{driver,total}}}{\dot{Q}_{\text{rated}}} \right\rceil. \quad (92)$$

Minimum number of elements to sustain standing losses without cycling:

$$N_{e,\text{min}} = \left\lceil \frac{\sum_{j=1}^{N_v} \dot{Q}_{\text{loss},j}}{\dot{Q}_{\text{rated}}} \right\rceil. \quad (93)$$

If the minimum stable driver output at turndown exceeds $\sum_j \dot{Q}_{\text{loss},j}$, the driver will cycle at low demand; $N_{e,\text{max}}$ should be increased or a smaller modulating element added.

J. Thermal Battery Sizing (Step 10)

1) Hydronic Priming Volume:

$$V_{\text{prime}} = V_{\text{pipe}} + V_{\text{coil,int}} + V_{\text{exp}}, \quad (94)$$

where, using the *internal* pipe diameter d_i :

$$V_{\text{coil,int}} = \frac{\pi d_i^2}{4} L_{\text{coil}}, \quad (95)$$

$$V_{\text{pipe}} = \sum_{k=1}^m \frac{\pi d_{i,k}^2}{4} \ell_k, \quad (96)$$

$$V_{\text{exp}} = f_{\text{exp}} (V_{\text{pipe}} + V_{\text{coil,int}}), \quad f_{\text{exp}} \approx 0.04\text{--}0.08. \quad (97)$$

Note the use of d_i here, in contrast to the d_o used in the displacement calculation (77). This is correct: the priming volume is the fluid payload inside the loop, not the volume it displaces.

2) Electric Element Flux Buffer Volume Formulation:

$$V_{\text{buffer,e}} = V_{\text{clearance}} + V_{\text{flux}}, \quad V_{\text{flux}} = \frac{\dot{Q}_e}{\rho c_p (dT/dt)_{\text{limit}}}, \quad (98)$$

where $(dT/dt)_{\text{limit}}$ is the maximum allowable temperature ramp rate that the SCR or VFD control system can track without triggering element high-temperature safety interlocks (K/s), and $V_{\text{clearance}}$ is the volume required to keep all active elements fully submerged at all times including at minimum standby fluid level.

3) *Fired Heater Minimum Volume from Burner Turndown Constraints:* To prevent the fluid temperature from rising through the controller deadband ΔT_{db} faster than the burner can throttle down during its minimum runtime $t_{\text{min,run}}$ at minimum stable output $\dot{Q}_{\text{min}} = \dot{Q}_{\text{rated}}/TR$:

$$V_{\text{min,fired}} = \frac{\dot{Q}_{\text{min}} t_{\text{min,run}}}{\rho c_p \Delta T_{\text{db}}}. \quad (99)$$

4) *Chiller Minimum Volume from Compressor Anti-Recycle Constraints:* The fluid temperature must not exceed $T_{\text{set}} + \Delta T_{\text{db}}$ during the compressor anti-recycle lockout period τ_{arc} :

$$V_{\text{min,chiller}} = \frac{\dot{Q}_{\text{loss}} \tau_{\text{arc}}}{\rho c_p \Delta T_{\text{db}}}. \quad (100)$$

5) Maximum Volume Derivation from First-Law Transient Energy Balance:

Governing ODE: Applying the first law to a well-mixed thermal battery of volume V with no driver input ($\dot{Q}_{\text{source}} = 0$):

$$\rho V c_p \frac{dT}{dt} = -\dot{Q}_{\text{loss}}(T) - \dot{Q}_{\text{proc}}, \quad (101)$$

where $\dot{Q}_{\text{loss}}(T) = (T - T_{\text{worst,amb}})/R_{\text{tot}}$ and $\dot{Q}_{\text{proc}} = Q_{\text{design}}/t_{\text{batch}}$ (constant peak extraction rate).

Integration and Autonomy Boundary: Under the conservative assumption $T \approx T_{\text{set}}$ (small temperature variation relative to absolute temperature), the total energy extracted during one batch cycle t_{batch} is:

$$\Delta E = (\dot{Q}_{\text{loss}} + \dot{Q}_{\text{proc}}) t_{\text{batch}}. \quad (102)$$

This must be provided entirely by cooling the battery through the allowable temperature drawdown $\Delta T_{\text{allow}} = T_{\text{max}} - T_{\text{min}}$:

$$\Delta E = \rho V c_p \Delta T_{\text{allow}}. \quad (103)$$

Equating (102) and (103):

$$V_{\text{max}} = \frac{(\dot{Q}_{\text{loss}} + \dot{Q}_{\text{proc}}) t_{\text{batch}}}{\rho c_p \Delta T_{\text{allow}} \eta_{\text{usable}}}, \quad (104)$$

where $\eta_{\text{usable}} = 0.80$ is the usable thermal fraction accounting for stratification and standing losses.

6) *Thermal Time Constant Formulation:* Setting $\dot{Q}_{\text{proc}} = 0$ in (101) and linearising the loss term, the natural cooling response is exponential with time constant:

$$\tau_{\text{th}} = \rho V c_p R_{\text{tot}}. \quad (105)$$

Equivalently, $\tau_{\text{th}} = (\rho V c_p)/\dot{Q}_{\text{loss}} \times (T_{\text{set}} - T_{\text{worst,amb}})$, confirming that a larger volume, better insulation (higher R_{tot}), and higher fluid thermal mass all increase autonomy as expected.

7) *Thermal Battery Tank Geometric Dimensioning Procedure*: The selected volume $V_{\text{batt}} \in [V_{\text{min}}, V_{\text{max}}]$ is converted to physical dimensions by applying the Step 3 cubic solution with V_{batt} in place of V_{target} and the same aspect ratio α and cap profile selected for the process vessel.

For the conical-capped case from (6):

$$d_{\text{batt}} = \left(\frac{24 V_{\text{batt}}}{\pi(6\alpha + \tan \theta)} \right)^{1/3}, \quad h_{2,\text{batt}} = \alpha d_{\text{batt}}. \quad (106)$$

For the hemi-ellipsoidal case, substitute into (11). The aspect ratio sanity bounds of Step 3 Section 3.2.1 apply: $h_{\text{total,batt}}/d_{\text{batt}} \in [0.8, 4.0]$. A minimum internal diameter of 400 mm is recommended for thermal battery tanks to ensure adequate element submersion and maintenance access.

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APPENDIX B THERMOPHYSICAL PROPERTY DATASETS AND REFERENCE MATERIAL

This appendix consolidates all baseline thermodynamic matrices, geometric line profiles, and lifecycle safety tolerances referenced across the core design framework. The tabulated data blocks listed herein provide the explicit property pairs required to execute the first-order linear interpolation algorithms detailed in Section A.A. Sections B.A through B.D serve as the unified data bank for optimizing calculations in Steps 2, 4, 5, 6, 8, and 9.

K. Aqueous and Industrial Transport Fluid Data Matrices

The following datasets represent the baseline thermodynamic properties of standard process and hydronic heat-transfer fluids. Designers must look up parameters at local bulk mean temperatures using first-order linear splines. All property values are evaluated at atmospheric pressure unless otherwise stated. For pressurised closed-loop systems, minor corrections to density and specific heat may be applied per the fluid manufacturer's pressurised property charts; however, for the operating envelopes covered by this framework, the atmospheric-baseline values introduce negligible error for sizing purposes.

1) *Pure Water Base Fluid (Process/Utility Standard)*: Table II indexes properties for distilled/purified water lines across atmospheric heating intervals, primarily utilised for process batch calculations and clean utility runs. Extended to 100°C at atmospheric pressure; pressurised systems may use water as a transport fluid beyond this limit with appropriate saturation-margin analysis.

TABLE II
THERMOPHYSICAL PROPERTIES OF PURE WATER (0°C TO 100°C)

Temp. T (°C)	Density ρ (kg/m ³)	Sp. Heat c_p (J/kg·K)	Dyn. Viscosity μ ($\times 10^{-3}$ Pa·s)	Kin. Viscosity ν ($\times 10^{-6}$ m ² /s)	Thermal Cond. k (W/m·K)	Prandtl Pr
0	999.8	4217	1.787	1.787	0.561	13.44
20	998.2	4182	1.002	1.004	0.598	7.01
40	992.2	4179	0.653	0.658	0.631	4.32
60	983.2	4185	0.467	0.475	0.654	2.99
80	971.8	4196	0.354	0.364	0.670	2.22
100	958.4	4216	0.282	0.294	0.679	1.75

2) *30% Propylene Glycol Aqueous Solution (Closed Loop Anti-Freeze Option)*: Table III provides metrics for commercial-grade inhibited propylene glycol mixtures at 30% by volume. Freeze point approximately −13°C. This fluid profile is common in HVAC and industrial process loops due to its low toxicity and GRAS (Generally Recognised As Safe) regulatory classification.

TABLE III
THERMOPHYSICAL PROPERTIES OF 30% PROPYLENE GLYCOL SOLUTION BY VOLUME

Temp. T (°C)	Density ρ (kg/m ³)	Sp. Heat c_p (J/kg·K)	Dyn. Viscosity μ ($\times 10^{-3}$ Pa·s)	Kin. Viscosity ν ($\times 10^{-6}$ m ² /s)	Thermal Cond. k (W/m·K)	Prandtl Pr
0	1042.0	3710	4.150	3.983	0.382	40.31
30	1035.0	3780	1.820	1.758	0.395	17.43
60	1029.0	3840	0.940	0.913	0.407	8.87
90	1025.0	3890	0.520	0.507	0.415	4.87
120	1014.0	3950	0.340	0.335	0.421	3.19

3) *50% Propylene Glycol Aqueous Solution (Heavy Freeze-Protection Standard)*: Table IV provides the industry-standard 50% by volume concentration, delivering a freeze point of approximately −33°C. This is the most commonly specified concentration for outdoor or unheated plant installations in temperate and cold climates. Viscosity penalty at low temperatures is significant; cold-start hydraulic loading must be evaluated against pump curve data.

TABLE IV
THERMOPHYSICAL PROPERTIES OF 50% PROPYLENE GLYCOL SOLUTION BY VOLUME

Temp. T (°C)	Density ρ (kg/m ³)	Sp. Heat c_p (J/kg·K)	Dyn. Viscosity μ ($\times 10^{-3}$ Pa·s)	Kin. Viscosity ν ($\times 10^{-6}$ m ² /s)	Thermal Cond. k (W/m·K)	Prandtl Pr
−20	1058.0	3570	22.400	21.172	0.358	223.2
0	1052.0	3620	8.600	8.175	0.366	85.0
20	1044.0	3670	3.800	3.640	0.374	37.3
40	1036.0	3730	2.050	1.979	0.381	20.1
60	1026.0	3790	1.210	1.179	0.387	11.8
80	1015.0	3850	0.780	0.769	0.391	7.7
100	1002.0	3910	0.530	0.529	0.394	5.3

4) *40% Ethylene Glycol Aqueous Solution (High-Efficiency HVAC/Chilling Loop)*: Table V catalogs industrial ethylene glycol parameters at 40% by volume, with a freeze point of approximately −24°C. Optimised for heavy-duty process chilling runs; note that ethylene glycol is toxic and requires appropriate secondary containment in food-adjacent environments.

5) *50% Ethylene Glycol Aqueous Solution (Maximum Standard Freeze Protection)*: Table VI documents the 50% by volume ethylene glycol concentration, delivering a freeze point of approximately −37°C. This is the maximum concentration recommended for standard hydronic service; higher concentrations offer diminishing freeze-point depression with increasing viscosity penalty. Widely stocked as a pre-mixed inhibited product.

TABLE V
THERMOPHYSICAL PROPERTIES OF 40% ETHYLENE GLYCOL SOLUTION BY VOLUME

Temp. T (°C)	Density ρ (kg/m ³)	Sp. Heat c_p (J/kg·K)	Dyn. Viscosity μ ($\times 10^{-3}$ Pa·s)	Kin. Viscosity ν ($\times 10^{-6}$ m ² /s)	Thermal Cond. k (W/m·K)	Prandtl Pr
−20	1078.0	3380	9.420	8.738	0.401	79.43
10	1063.0	3480	2.860	2.691	0.418	23.81
40	1049.0	3580	1.340	1.277	0.429	11.18
70	1033.0	3680	0.790	0.765	0.438	6.64
100	1016.0	3790	0.510	0.502	0.443	4.36

TABLE VI
THERMOPHYSICAL PROPERTIES OF 50% ETHYLENE GLYCOL SOLUTION BY VOLUME

Temp. T (°C)	Density ρ (kg/m ³)	Sp. Heat c_p (J/kg·K)	Dyn. Viscosity μ ($\times 10^{-3}$ Pa·s)	Kin. Viscosity ν ($\times 10^{-6}$ m ² /s)	Thermal Cond. k (W/m·K)	Prandtl Pr
−30	1089.0	3260	27.600	25.344	0.384	234.4
−10	1079.0	3310	7.400	6.858	0.393	62.4
10	1069.0	3360	3.280	3.068	0.401	27.5
30	1057.0	3420	1.870	1.769	0.408	15.7
60	1039.0	3510	0.980	0.943	0.416	8.3
90	1019.0	3610	0.590	0.579	0.421	5.1

6) *Therminol 66 (Synthetic Organic High-Temperature Thermal Oil Standard)*: Table VII documents the behaviour of Therminol 66, a synthetic organic heat-transfer liquid manufactured by Eastman. Applied when hot-side temperatures must exceed 150°C to 345°C without generating extreme steam pressures. Note the extreme Prandtl number at cold-start conditions (50°C): the cold-start viscosity correction for U_{cold} in Step 6 is mandatory for this fluid class.

TABLE VII
THERMOPHYSICAL PROPERTIES OF THERMINOL 66 LIQUID HEAT TRANSFER FLUID

Temp. T (°C)	Density ρ (kg/m ³)	Sp. Heat c_p (J/kg·K)	Dyn. Viscosity μ ($\times 10^{-3}$ Pa·s)	Kin. Viscosity ν ($\times 10^{-6}$ m ² /s)	Thermal Cond. k (W/m·K)	Prandtl Pr
50	981.0	1660	58.200	59.327	0.116	832.74
100	953.0	1845	6.800	7.135	0.112	111.96
150	923.0	2030	1.950	2.113	0.108	36.65
200	893.0	2215	0.880	0.985	0.104	18.74
250	861.0	2400	0.490	0.569	0.101	11.64
300	828.0	2585	0.310	0.374	0.097	8.26
345	795.0	2760	0.210	0.264	0.093	6.23

7) *Therminol VP-1 / Dowtherm A (Diphenyl Oxide Eutectic, High-Performance Industrial Standard)*: Table VIII covers Therminol VP-1 and its equivalent Dowtherm A (both are diphenyl/diphenyl oxide eutectic blends with near-identical property profiles). Operating range 12°C to 400°C. This is the primary benchmark fluid for solar thermal, concentrated chemical processing, and high-temperature reactor jacket applications. Vapour pressure is non-negligible above 200°C; system pressurisation must be confirmed against saturation data. Do not use in food-contact environments.

TABLE VIII
THERMOPHYSICAL PROPERTIES OF THERMINOL VP-1 / DOWTHERM A

Temp. T (°C)	Density ρ (kg/m ³)	Sp. Heat c_p (J/kg·K)	Dyn. Viscosity μ ($\times 10^{-3}$ Pa·s)	Kin. Viscosity ν ($\times 10^{-6}$ m ² /s)	Thermal Cond. k (W/m·K)	Prandtl Pr
20	1060.0	1550	3.990	3.764	0.138	44.81
50	1039.0	1600	1.880	1.809	0.135	22.27
100	1004.0	1680	0.940	0.936	0.130	12.15
150	968.0	1760	0.560	0.579	0.125	7.88
200	930.0	1850	0.370	0.398	0.119	5.75
250	891.0	1950	0.260	0.292	0.113	4.49
300	849.0	2060	0.190	0.224	0.107	3.66
350	805.0	2180	0.145	0.180	0.101	3.13
400	757.0	2310	0.112	0.148	0.095	2.73

8) *Caloria HT 43 (Mineral-Based Thermal Oil, Low-to-Medium Temperature)*: Table IX documents Caloria HT 43 (ExxonMobil), a highly refined mineral oil thermal fluid. Operating range −10°C to 170°C. Lower procurement cost than synthetic alternatives makes it attractive for budget-constrained installations with moderate temperature requirements. Oxidation stability is inferior to synthetic fluids; fluid monitoring and periodic replacement schedules are mandatory above 140°C. Not food-grade.

9) *Mobiltherm 603 (Mineral-Based Thermal Oil, Standard Industrial Grade)*: Table X covers Mobiltherm 603 (ExxonMobil Mobil), a widely distributed mineral thermal oil for continuous service up to 315°C. One of the most commonly stocked thermal oils globally, making it the practical default for installations where supply chain reliability is prioritised over peak thermal performance. Flash point $\approx$ 180°C; fire safety classification must be confirmed against local plant codes.

10) *Paratherm NF (Food-Safe Synthetic Thermal Oil, NSF HX-1 Registered)*: Table XI documents Paratherm NF, a synthetic paraffinic thermal fluid carrying NSF HX-1 registration for incidental food contact. Operating range −20°C to 260°C. This

TABLE IX
THERMOPHYSICAL PROPERTIES OF CALORIA HT 43 MINERAL THERMAL OIL

Temp. T (°C)	Density ρ (kg/m ³)	Sp. Heat c_p (J/kg·K)	Dyn. Viscosity μ ($\times 10^{-3}$ Pa·s)	Kin. Viscosity ν ($\times 10^{-6}$ m ² /s)	Thermal Cond. k (W/m·K)	Prandtl Pr
20	869.0	1950	43.500	50.058	0.134	633.2
40	857.0	2000	18.200	21.237	0.131	278.0
60	845.0	2060	8.600	10.178	0.129	137.4
80	833.0	2120	4.500	5.402	0.127	75.1
100	820.0	2180	2.600	3.171	0.125	45.3
130	800.0	2270	1.250	1.563	0.122	23.3
160	780.0	2360	0.680	0.872	0.119	13.5

TABLE X
THERMOPHYSICAL PROPERTIES OF MOBILTHERM 603 THERMAL OIL

Temp. T (°C)	Density ρ (kg/m ³)	Sp. Heat c_p (J/kg·K)	Dyn. Viscosity μ ($\times 10^{-3}$ Pa·s)	Kin. Viscosity ν ($\times 10^{-6}$ m ² /s)	Thermal Cond. k (W/m·K)	Prandtl Pr
40	868.0	1980	16.800	19.355	0.133	250.2
80	844.0	2080	4.750	5.629	0.129	76.6
120	820.0	2190	1.920	2.341	0.125	33.6
160	795.0	2300	0.970	1.220	0.121	18.4
200	769.0	2420	0.570	0.741	0.117	11.8
250	736.0	2570	0.330	0.448	0.112	7.6
300	702.0	2720	0.210	0.299	0.107	5.3

is the preferred selection for food processing, pharmaceutical manufacturing, and confectionery applications where regulatory compliance requires a certified food-safe fluid but the operating temperature exceeds the practical range of glycol-water solutions. Thermally stable with low odour and low toxicity.

TABLE XI
THERMOPHYSICAL PROPERTIES OF PARATHERM NF FOOD-SAFE THERMAL OIL

Temp. T (°C)	Density ρ (kg/m ³)	Sp. Heat c_p (J/kg·K)	Dyn. Viscosity μ ($\times 10^{-3}$ Pa·s)	Kin. Viscosity ν ($\times 10^{-6}$ m ² /s)	Thermal Cond. k (W/m·K)	Prandtl Pr
20	840.0	2040	10.700	12.738	0.140	155.9
50	823.0	2100	4.100	4.983	0.137	62.8
80	806.0	2170	1.980	2.457	0.134	32.1
120	781.0	2270	0.880	1.127	0.130	15.4
160	756.0	2370	0.470	0.622	0.126	8.8
200	729.0	2470	0.290	0.398	0.122	5.9
240	701.0	2580	0.195	0.278	0.118	4.3

11) *Syltherm 800 (Silicone-Based Thermal Fluid, Wide-Range Specialty)*: Table XII covers Syltherm 800 (Dow), a polydimethylsiloxane-based (silicone) thermal fluid rated for continuous service from -40°C to 260°C , with intermittent use to 315°C . Outstanding low-temperature fluidity (low pour point, moderate viscosity at -40°C) makes it the dominant selection for pharmaceutical freeze-dry jacket loops, semiconductor temperature conditioning, and wide-range process cycling. Higher cost than hydrocarbon alternatives; silicone fluid contamination is incompatible with certain downstream processes and coating operations.

TABLE XII
THERMOPHYSICAL PROPERTIES OF SYLTHERM 800 SILICONE THERMAL FLUID

Temp. T (°C)	Density ρ (kg/m ³)	Sp. Heat c_p (J/kg·K)	Dyn. Viscosity μ ($\times 10^{-3}$ Pa·s)	Kin. Viscosity ν ($\times 10^{-6}$ m ² /s)	Thermal Cond. k (W/m·K)	Prandtl Pr
-40	1006.0	1380	21.000	20.874	0.132	219.5
-20	990.0	1420	8.700	8.788	0.130	95.1
0	975.0	1460	4.200	4.308	0.128	47.9
40	943.0	1540	1.600	1.697	0.124	19.9
80	910.0	1620	0.800	0.879	0.120	10.8
120	877.0	1710	0.470	0.536	0.116	6.9
160	843.0	1800	0.310	0.368	0.112	5.0
200	808.0	1900	0.220	0.272	0.108	3.9
260	756.0	2050	0.145	0.192	0.102	2.9

12) *50% Glycerin Aqueous Solution (Sanitary Food / Pharmaceutical Standard)*: Table XIII provides metrics for refined aqueous glycerol loops at 50% by mass. This choice is common in sanitary process facilities due to its bio-safe, non-toxic profile under accidental product-contact failure scenarios. Notably high viscosity demands careful pump selection and hydraulic loop analysis; laminar flow conditions are likely at lower operating temperatures and must be evaluated against the Nusselt number correlations appropriate for the flow regime.

13) *Food-Grade Vegetable Thermal Oil (Confectionery / Edible Oil Processing)*: Table XIV provides representative thermophysical data for refined food-grade vegetable oil (palm/sunflower/canola blend), applicable to confectionery, chocolate processing, edible oil refining, and dairy fat handling installations where direct product contact is a credible failure mode. Properties are presented for a representative refined palm-sunflower blend; actual values vary with specific oil composition and should be

TABLE XIII
THERMOPHYSICAL PROPERTIES OF 50% GLYCERIN SOLUTION BY MASS

Temp. T (°C)	Density ρ (kg/m ³)	Sp. Heat c_p (J/kg·K)	Dyn. Viscosity μ ($\times 10^{-3}$ Pa·s)	Kin. Viscosity ν ($\times 10^{-6}$ m ² /s)	Thermal Cond. k (W/m·K)	Prandtl Pr
10	1134.0	3220	7.950	7.011	0.415	61.68
30	1124.0	3300	3.820	3.399	0.428	29.45
50	1113.0	3390	2.180	1.959	0.439	16.84
70	1101.0	3480	1.390	1.262	0.446	10.85
90	1088.0	3570	0.960	0.882	0.450	7.62

verified against the supplier's data sheet for the procurement batch. Operating ceiling approximately 180°C; oxidation degradation accelerates above 150°C and fluid replacement intervals must be shortened accordingly.

TABLE XIV
THERMOPHYSICAL PROPERTIES OF FOOD-GRADE VEGETABLE THERMAL OIL

Temp. T (°C)	Density ρ (kg/m ³)	Sp. Heat c_p (J/kg·K)	Dyn. Viscosity μ ($\times 10^{-3}$ Pa·s)	Kin. Viscosity ν ($\times 10^{-6}$ m ² /s)	Thermal Cond. k (W/m·K)	Prandtl Pr
40	910.0	1950	36.000	39.560	0.168	417.9
60	898.0	1990	16.200	18.040	0.166	194.4
80	886.0	2030	8.500	9.594	0.164	105.2
100	873.0	2080	4.900	5.613	0.162	62.9
120	860.0	2130	3.100	3.605	0.160	41.3
150	840.0	2200	1.700	2.024	0.157	23.8

14) *D'Limonene-Based Thermal Fluid (Bio-Derived, Food-Adjacent Specialty)*: Table XV documents thermal fluid properties for high-purity d-limonene, a bio-derived citrus terpene increasingly specified in food-adjacent installations due to its natural origin, low toxicity, and pleasant odour profile. Operating range approximately 0°C to 175°C. While thermophysical performance is modest relative to synthetic thermal oils, regulatory acceptance in food-contact risk environments and lower environmental impact make it a commercially viable niche selection. Flash point $\approx$ 49°C; fire classification is Class IC flammable liquid and strict hot-work permit controls apply.

TABLE XV
THERMOPHYSICAL PROPERTIES OF D'LIMONENE BIO-DERIVED THERMAL FLUID

Temp. T (°C)	Density ρ (kg/m ³)	Sp. Heat c_p (J/kg·K)	Dyn. Viscosity μ ($\times 10^{-3}$ Pa·s)	Kin. Viscosity ν ($\times 10^{-6}$ m ² /s)	Thermal Cond. k (W/m·K)	Prandtl Pr
20	841.0	1860	0.920	1.094	0.143	11.97
40	828.0	1910	0.730	0.882	0.140	9.97
60	814.0	1960	0.590	0.725	0.138	8.38
80	800.0	2010	0.490	0.613	0.135	7.29
100	786.0	2060	0.410	0.522	0.133	6.35
140	756.0	2170	0.300	0.397	0.128	5.09

L. Solid Material and Insulation Thermal Properties

Conductive resistances (R_{wall} and $R_{insulation}$) across Steps 6 and 9 rely on core material transport metrics. Table XVI establishes wall conductivities, while Table XVII provides insulation property targets evaluated at 20°C atmospheric reference settings.

TABLE XVI
THERMAL CONDUCTIVITIES OF STANDARD STRUCTURAL METAL ALLOYS

Metallurgical Material Specification	Thermal Conductivity k (W/m·K)
Carbon Steel (AISI 1020 / Standard Line)	50.0
Stainless Steel 304 (Austenitic Standard)	16.2
Stainless Steel 316 (High-Corrosion Resist)	15.0
Nickel Alloy 625 (Inconel Core Variant)	9.8
Copper (Drawn Tube Standard C12200)	390.0
Titanium (Grade 2 Unalloyed Core)	21.9

TABLE XVII
THERMAL PROPERTIES OF INDUSTRIAL SHELL INSULATION CLADDINGS

Insulation Material Type	Thermal Conductivity k (W/m·K)	Max Temp Rating (°C)
Rigid Polyurethane Foam (PUF)	0.024	120
Mineral Wool / Rockwool Board	0.038	650
Fiberglass Blanket Insulation	0.040	540
Expanded Polystyrene (EPS)	0.035	75
Calcium Silicate Structural Sheet	0.055	900

M. Industrial Design Boundary Allowances

To ensure the sized machinery retains performance margins across multi-year operational cycles, fluid friction and thermal performance elements must be penalised using surface degradation metrics.

1) *Absolute Pipe Material Surface Roughness Values:* Table XVIII specifies the absolute physical surface roughness (ε) used as the core input seed parameter for the Colebrook-White iterative hydraulic head loss calculations (Step 5).

TABLE XVIII
ABSOLUTE SURFACE ROUGHNESS PARAMETERS (ε)

Piping Material Specification Profile	Absolute Roughness ε (mm)
Drawn Plastic Piping (PVC / CPVC / PVDF)	0.0015
Drawn Copper Tubing (Clean Commercial)	0.0015
Commercial Stainless Steel Line (New/Clean)	0.0150
Commercial Carbon Steel / Wrought Iron	0.0450
Galvanized Iron / Zinc Coated Steel	0.1500
Heavily Corroded / Scaled Cast Iron Shell	1.5000

2) *Standard Industrial Fluid Fouling Heat Transfer Resistances:* Table XIX logs the standard thermal fouling resistance penalties (R_{f_in} , R_{f_out}). These terms apply localised heat-blocking penalties to the series thermal network calculation (Step 6) to model scaling and process deposition across the lifespan of the equipment.

TABLE XIX
STANDARD INDUSTRIAL FOULING THERMAL RESISTANCE PENALTIES

Fluid Circuit Operating Environment	Fouling Resistance R_f ($\text{m}^2 \cdot \text{K}/\text{W}$)
Distilled Water / Demineralised Loop Fluid	0.0001
Treated Boiler Feedwater (Closed Utility Run)	0.0002
Clean Inhibited Glycol Aqueous Solutions	0.0002
City Tap Water (Standard Municipal Intake)	0.0003
River Water / Untreated Cooling Tower Fluid	0.0005
Light Organic Process Fractions / Light Oils	0.0004
Synthetic Thermal Oils (Clean Service)	0.0002
Mineral Thermal Oils (Standard Service)	0.0003
Silicone Thermal Fluids (Clean Service)	0.0001
Heavy Crude Fractions / Slurry Process Fluids	0.0010

APPENDIX C COMMERCIAL PIPING STANDARDS AND MATERIAL PROFILES

This appendix consolidates the mechanical, geometric, and thermomechanical data required to execute the Nominal Pipe Size (NPS) selection and structural material confirmation outlined in Step 4 of the framework. It includes standardized ASME dimensional boundaries for fluid flow cross-sections and comprehensive material evaluation matrices to ensure thermodynamic efficiency, structural integrity, and chemical compatibility across the operational lifespan of the hydronic system.

N. ASME B36.10M and B36.19M Pipe Dimensional Data

To safely constrain internal fluid velocities below the erosion ceiling ($v \leq v_{max}$), the theoretical minimum flow area calculated in Step 4 must be matched to a discrete, commercially available pipe profile. Table XX lists standard dimensions for Welded and Seamless Wrought Steel (ASME B36.10M) and Stainless Steel (ASME B36.19M) pipe across typical industrial schedules.

TABLE XX
STANDARD COMMERCIAL PIPE DIMENSIONS (NPS 1/2" TO 8")

Nominal Size NPS (inches)	Outer Diameter d_{outer} (mm)	Schedule Designator	Wall Thickness t_{wall} (mm)	Inner Diameter d_{in} (mm)	Internal Flow Area A_{in} ($\times 10^{-3}$ m ²)
1/2"	21.34	Sch 10S	2.11	17.12	0.230
	21.34	Sch 40	2.77	15.80	0.196
	21.34	Sch 80	3.73	13.88	0.151
1"	33.40	Sch 10S	2.77	27.86	0.610
	33.40	Sch 40	3.38	26.64	0.557
	33.40	Sch 80	4.55	24.30	0.464
1-1/2"	48.26	Sch 10S	2.77	42.72	1.433
	48.26	Sch 40	3.68	40.90	1.314
	48.26	Sch 80	5.08	38.10	1.140
2"	60.33	Sch 10S	2.77	54.79	2.358
	60.33	Sch 40	3.91	52.51	2.165
	60.33	Sch 80	5.54	49.25	1.905
3"	88.90	Sch 10S	3.05	82.80	5.384
	88.90	Sch 40	5.49	77.92	4.768
	88.90	Sch 80	7.62	73.66	4.261
4"	114.30	Sch 10S	3.05	108.20	9.195
	114.30	Sch 40	6.02	102.26	8.213
	114.30	Sch 80	8.56	97.18	7.417
6"	168.28	Sch 10S	3.40	161.48	20.480
	168.28	Sch 40	7.11	154.06	18.641
	168.28	Sch 80	10.97	146.34	16.819
8"	219.08	Sch 10S	3.76	211.56	35.153
	219.08	Sch 40	8.18	202.72	32.276
	219.08	Sch 80	12.70	193.68	29.462

O. Thermomechanical Material Profiles

Selecting the correct material for the vessel shell and internal helical coil directly impacts the Overall Heat Transfer Coefficient (U) derived in Step 6, as well as the maximum structural limits of the system. Table XXI indexes the baseline mechanical and thermal properties of standard industrial materials.

TABLE XXI
THERMOMECHANICAL PROPERTIES OF COMMERCIAL STRUCTURAL & PIPING MATERIALS

Material Specification (Common Designation)	Density ρ (kg/m ³)	Thermal Cond. k (W/m · K)	Yield Strength S_y (MPa)	Max Operating Temp (°C)	Relative Cost Factor
<i>Ferrous Alloys (Standard Vessel & Structural Shells)</i>					
Carbon Steel (AISI 1020)	7850	50.0	250	425	1.0x
Carbon Steel (A106 Grade B)	7850	48.0	240	400	1.1x
Cast Iron (Gray, Class 30)	7150	45.0	130	350	0.8x
<i>Stainless Alloys (Sanitary & Corrosive Environments)</i>					
Stainless Steel (304 / 304L)	8000	16.2	215	870	3.5x
Stainless Steel (316 / 316L)	8000	15.0	205	870	4.5x
Duplex Stainless (2205)	7800	19.0	450	300	6.0x
<i>Non-Ferrous & Specialty Metals (High Thermal Transfer)</i>					
Copper (Drawn, C12200)	8940	390.0	100	200	4.0x
Aluminum (6061-T6)	2700	167.0	275	150	2.0x
Titanium (Grade 2 Unalloyed)	4510	21.9	275	315	12.0x
Inconel (Alloy 625)	8440	9.8	460	980	15.0x
<i>Industrial Polymers (Low-Temp Chilling & Utility)</i>					
PVC (Polyvinyl Chloride)	1380	0.19	40	60	0.5x
CPVC (Chlorinated PVC)	1550	0.14	55	93	0.8x
PEX (Cross-linked Polyethylene)	940	0.40	15	95	0.7x
PTFE (Teflon / Lined Piping)	2200	0.25	25	260	8.0x

P. Material Compatibility and Selection Criteria Matrix

To prevent galvanic corrosion, chemical degradation, or thermal failure, the selected piping and vessel materials must be validated against the primary process fluid and the hydronic heat-transfer fluid. Table XXII provides a selection matrix mapping standard materials to their appropriate operational environments.

TABLE XXII
MATERIAL SUITABILITY MATRIX FOR HYDRONIC PROCESS FLUIDS

Hydronic / Process Fluid	Carbon Steel	SS 304 / 316	Copper	CPVC / PEX
Pure Water (Deionized / RO)	POOR (Rusts)	EXCELLENT	GOOD	EXCELLENT
Treated Boiler Feedwater (Hot)	EXCELLENT	EXCELLENT	GOOD	POOR (Temp Limits)
Propylene / Ethylene Glycols	GOOD (w/ Inhibitor)	EXCELLENT	EXCELLENT	GOOD (Low Temp)
Therminol 66 (Thermal Oil)	EXCELLENT	EXCELLENT	POOR	UNSUITABLE
Glycerin Solutions (Sanitary)	POOR (Taints)	EXCELLENT	POOR	POOR
Brine / High-Chloride Chilling	POOR (Pitting)	POOR (Requires 316)	GOOD	EXCELLENT

Note: "EXCELLENT" denotes standard industry application with lifespan > 15 years.

"POOR" indicates rapid degradation, galvanic potential, or unacceptable process contamination.

C.D Pipe Fittings and Connections

The hydraulic resistance characterisation performed in Step 5 of the framework requires that all fittings installed within the hydronic circuit be accounted for through their equivalent pipe length contributions. Each fitting introduces a localised pressure drop that, if neglected, will cause the pump selection to be undersized and the actual flow velocity to fall below design intent. Table XXIII provides standard equivalent length values (L_{eq}) for common wrought and forged fittings conforming to ASME B16.9 (butt-weld), ASME B16.11 (socket-weld and threaded), and ASME B16.5 (flanges). These values are expressed as a ratio to nominal pipe diameter (L_{eq}/d_{in}) and are valid for fully turbulent flow conditions consistent with coil operating Reynolds numbers.

TABLE XXIII
EQUIVALENT LENGTH RATIOS FOR STANDARD INDUSTRIAL FITTINGS (ASME B16.9 / B16.11)

Fitting Type	Connection	Equivalent Length Ratio L_{eq}/d_{in}		
	Standard	NPS $\leq 1''$	NPS 1.5''–3''	NPS 4''–8''
<i>Elbows</i>				
90° Standard Elbow	B16.9 BW	30	30	30
90° Long-Radius Elbow	B16.9 BW	16	16	16
45° Standard Elbow	B16.9 BW	16	16	16
<i>Tees</i>				
Tee, Flow-Through (Run)	B16.9 BW	20	20	20
Tee, Flow-Through (Branch)	B16.9 BW	60	60	60
<i>Reducers & Couplings</i>				
Concentric Reducer	B16.9 BW	5	5	5
Eccentric Reducer	B16.9 BW	5	5	5
Full-Coupling / Union	B16.11 SW	2	2	2
<i>Valves (Fully Open)</i>				
Gate Valve	B16.10	8	8	8
Globe Valve	B16.10	340	340	340
Ball Valve (Full Port)	B16.34	3	3	3
Butterfly Valve	B16.34	45	45	45
Check Valve (Swing)	B16.10	100	100	100
<i>Flanged Connections</i>				
Weld-Neck Flange (pair)	B16.5	2	2	2
Slip-On Flange (pair)	B16.5	2	2	2
<i>Note: BW = Butt-Weld; SW = Socket-Weld. Values based on Crane TP-410 turbulent flow data. For laminar flow ($Re < 2300$), apply the laminar correction factor $K_L = K \cdot (Re/1000)$.</i>				

C.E Welding and Joining Standards

All pressure-retaining joints in the hydronic vessel and coil assembly must conform to a recognised fabrication code. The applicable standard is determined by the operating pressure class, material group, and service fluid. Table XXIV maps common joint configurations to their governing ASME and AWS codes and specifies the minimum examination requirements that must be satisfied before the design verification gate in Step 11. Compliance with these requirements is a prerequisite for the formal sign-off on the completed design package.

TABLE XXIV
WELDING AND JOINING STANDARDS FOR HYDRONIC PRESSURE VESSELS AND COIL ASSEMBLIES

Joint Application	Governing Code	Weld Process	Filler / Electrode	Min. Examination Req.
Carbon Steel Vessel Shell (Butt Weld)	ASME Sec. IX / ASME VIII Div. 1	SMAW, GTAW, GMAW	E7018 / ER70S-6	Visual + RT or UT per UW-11
SS 304/316 Vessel Shell	ASME Sec. IX	GTAW (Root), GMAW (Fill)	ER308L / ER316L	Visual + 100% PT on root pass
Coil-to-Header Connection	ASME B31.3 (Process)	GTAW (orbital preferred)	ER316L / ER308L	Visual + 5% RT spot check
Coil Butt Joints (Same Material)	ASME B31.3	GTAW	Match base metal	Visual + Hydrostatic proof test
Nozzle / Stub-In (Vessel Shell)	ASME VIII Div. 1 UW-16	SMAW / GTAW	Match base metal	Visual + MT or PT (full)
Dissimilar Metal (CS to SS)	ASME Sec. IX / AWS D1.6	GTAW (butter layer)	ENiCrFe-3 (Inconel)	Visual + 100% RT
Flange Face (Slip-On / Weld-Neck)	ASME B16.5 / Sec. IX	SMAW / GTAW	Match base metal	Visual + hydrostatic leak test
Low-Pressure (< 150 psig)	Utility AWS D1.1 (Structural)	SMAW / FCAW	E7018 / E71T-1	Visual per AWS D1.1 Table 6.1

RT = Radiographic Testing; UT = Ultrasonic Testing; PT = Liquid Penetrant Testing; MT = Magnetic Particle Testing. All welders must hold current ASME Section IX or AWS QC1 qualification for the applicable process and material group.

C.F Gasket and Seal Material Compatibility

Flanged connections and inspection ports within the hydronic circuit require a gasket or seal whose material is independently verified against both the process fluid and the operating temperature range. Selection errors at this stage are a common root cause of slow seepage failures that are not captured during short-duration hydrostatic proof tests. Table XXV maps standard gasket and elastomeric seal materials to their effective operational envelopes. The temperature ratings listed represent continuous service limits; cyclic thermal shock conditions typical of batch processing duty should be assessed with a 15% de-rating applied to the upper temperature boundary.

TABLE XXV
GASKET AND SEAL MATERIAL COMPATIBILITY FOR HYDRONIC PROCESS FLUIDS

Gasket / Seal Material	Max (°C)	Temp.	Max (bar)	Press.	Compatible Fluids	Incompatible / Caution
PTFE (Virgin / Expanded)	260		70		Water, glycols, glycerin, acids, thermal oils, brine	Strong oxidisers; PTFE cold-flows under bolt load — use spiral-wound type above 40 bar
Spiral-Wound SS/PTFE	450		250		Steam, hot water, glycols, thermal oils	Not for cryogenic; requires flat-face flange
Nitrile (NBR)	120		35		Water, glycols, mineral oils, brine	Aromatic hydrocarbons; thermal oils (Therminol) cause swelling
EPDM (Ethylene Propylene)	150		20		Hot water, steam (low press.), glycols, dilute acids	Petroleum-based oils and solvents; Therminol series
Viton (FKM Fluoroelastomer)	200		35		Thermal oils, glycols, brine, mild acids, hot water	Steam above 150°C; ketones; low-molecular-weight esters
Graphite (Flexible Foil)	450		200		Hot water, steam, thermal oils, glycols, most process fluids	Strong oxidising acids (e.g. HNO ₃); not food-grade without encapsulation
Neoprene (CR)	100		20		Water, glycols, brine, mild alkalis	Aromatic solvents; thermal oils; strong oxidisers
Silicone (VMQ)	230		15		Hot water (low press.), steam (low press.), food-grade glycerin	Petroleum oils; acids; steam above 150°C

*All food-contact or sanitary applications must use FDA-compliant (21 CFR 177) grades of PTFE, EPDM, or Silicone.
Spiral-wound gaskets must be specified with inner and outer guide rings per ASME B16.20.*

C.G Insulation Materials and Thermal Properties

The environmental heat loss term Q_{loss} quantified in Step 5 of the framework is governed by the total thermal resistance of the vessel boundary layer, which includes the insulation jacket as the dominant resistance element. The selection of insulation type directly affects both the steady-state balance load and the payback period on the installed thermal capacity. Table XXVI catalogues the thermal and mechanical properties of industrial insulation systems appropriate for hydronic vessel jacketing across the operating temperature range of the framework. The effective thermal conductivity values (k) listed are mean-temperature values; for precision loss calculations, the temperature-dependent $k(T)$ curve from the manufacturer's datasheet should be used to integrate across the wall temperature gradient.

TABLE XXVI
THERMAL AND MECHANICAL PROPERTIES OF INDUSTRIAL VESSEL INSULATION MATERIALS

Insulation Material	Thermal Cond. k (W/m·K)	Density ρ (kg/m ³)	Service Temp. Range (°C)	Relative Cost	Typical Application
<i>Rigid Foam Systems</i>					
Polyurethane Foam (PUR)	0.022–0.028	30–60	–50 to +120	1.0x	Cold tanks, chilling vessels
Polyisocyanurate (PIR)	0.022–0.026	32–48	–50 to +150	1.2x	Heated vessels, pipework
Extruded Polystyrene (XPS)	0.030–0.038	25–45	–50 to +75	0.7x	Low-temp utility; moisture-resistant
<i>Mineral Fibre Systems</i>					
Glass Wool (Wired Mat)	0.032–0.044	10–48	Up to +450	0.6x	General industrial pipework
Rock Wool (Stone Wool)	0.033–0.045	40–200	Up to +750	0.8x	High-temp vessels and steam lines
Calcium Silicate Board	0.055–0.085	190–300	Up to +1000	1.5x	Very high-temp, structural duty
<i>Flexible & Specialty Systems</i>					
Elastomeric Foam (Armaflex)	0.033–0.040	60–80	–50 to +105	1.8x	Cold pipework; self-sealing joints
Aerogel Blanket	0.013–0.018	100–200	–200 to +650	8.0x	Space-constrained high-value duty
Cellular Glass (Foamglas)	0.038–0.055	100–130	–265 to +430	3.5x	Cryogenic and high-humidity service
<i>Cladding and Jacketing (Outer Weather Barrier)</i>					
Aluminium Sheet (0.6–1.0 mm)	N/A	2700	–200 to +200	1.0x	Standard outdoor and indoor jacketing
Stainless Sheet (0.5–0.8 mm)	N/A	8000	All ranges	2.5x	Corrosive or sanitary environments
PVC Jacketing (Vapour Barrier)	N/A	1380	Up to +60	0.5x	Indoor low-temp chilling only

For the VCH loss model, minimum recommended insulation thickness targets $R_{ins} \geq 1.0 \text{ m}^2 \cdot \text{K/W}$ for heated vessels.

Insulation systems in food-processing or pharmaceutical environments must comply with local hygiene codes (e.g. EHEDG, 3-A Sanitary Standards).

C.H Coil Fabrication Tolerances and Bend Radius Data

The coil geometry optimisation performed in Step 7 produces a theoretical helix diameter, coil pitch, and number of turns. These ideal dimensions must be mapped to physically realisable fabrication parameters before the coil can be ordered or manufactured. Two constraints govern this translation: the minimum bend radius that a given pipe size and material can sustain without wall thinning or ovality beyond acceptable limits, and the compound packing efficiency factor η_{pack} that accounts for the systematic loss of usable coil turns introduced by fabrication tolerances, structural supports, weld access clearances, thermal expansion gaps, and dead-zone prevention offsets at the top and bottom of the vessel.

Table XXVII provides the minimum centreline bend radius (R_{min}) expressed as a multiple of nominal outer diameter (D) for mandrel bending of seamless and welded tube, consistent with ASME B16.49 (factory bends) and EN 13480-4 (industrial piping). Table XXVIII quantifies the individual tolerance deductions that compound into the η_{pack} correction applied in Step 7.

TABLE XXVII
MINIMUM CENTRELINE BEND RADIUS FOR HELICAL COIL FABRICATION

Pipe Material	Fabrication Method	Min. Centreline Radius R_{min} (multiples of D_{outer})			
		NPS 1/2"–1"	NPS 1.5"–2"	NPS 3"–4"	NPS 6"–8"
Carbon Steel (Seamless)	Cold Mandrel Bend	$1.5D$	$2.0D$	$3.0D$	$5.0D$
Carbon Steel (Seamless)	Hot Induction Bend	$1.5D$	$1.5D$	$2.0D$	$3.0D$
Stainless 304/316 (Seamless)	Cold Mandrel Bend	$2.0D$	$2.5D$	$3.5D$	$5.0D$
Stainless 304/316 (Seamless)	Hot Induction Bend	$1.5D$	$2.0D$	$2.5D$	$3.5D$
Copper (Drawn, C12200)	Cold Spring Bending	$1.5D$	$2.0D$	$3.0D$	N/A
Titanium Grade 2	Cold Mandrel Bend	$3.0D$	$3.5D$	$5.0D$	N/A

*Radii below these limits risk wall thinning > 12.5% or ovality > 8% per ASME B16.49 acceptance criteria.
For tight-pitch helical coils wound on a mandrel, confirm that the helix diameter $D_{helix} \geq 2 \times R_{min}$.*

TABLE XXVIII
COIL PACKING EFFICIENCY DEDUCTION FACTORS COMPRISING η_{pack}

Deduction Source	Physical Basis	Typical Deduction	Notes
Structural Support Legs	Radial supports interrupt coil continuity at ≥ 3 points	2–4 turns	Fixed loss per coil
Weld Access Clearance (Top)	Minimum 50–75 mm clearance from vessel top head to first active turn	1–2 turns	Increases with NPS
Weld Access Clearance (Bottom)	Minimum 50–75 mm clearance from vessel base to last active turn	1–2 turns	Increases with NPS
Thermal Expansion Gap	Axial growth of helical coil at $\Delta T_{operating}$	0.5–1 turn	Per 10 m coil length
Dead-Zone Prevention (Top)	Stagnant fluid zone above coil top must be minimised	1 turn	Heating applications
Dead-Zone Prevention (Bottom)	Stagnant fluid zone below coil bottom must be minimised	1 turn	Cooling applications
Coil Pitch Rounding	Discrete turn count rounded down to nearest integer	0–1 turn	Deterministic

Typical Composite η_{pack} Range **0.80 – 0.92**

*η_{pack} is applied as: $N_{active} = \lfloor N_{theoretical} \times \eta_{pack} \rfloor$, where $N_{theoretical}$ is derived from the Step 7 optimisation.
For large-diameter vessels ($ID > 1.5$ m) with multiple coil banks, each bank should carry an independent η_{pack} assessment.*

APPENDIX D

DISTRIBUTED PROCESS AUTOMATION AND CONTROL ARCHITECTURE

This appendix defines the system-level control philosophy, instrumentation topology, and automation sequencer governing the multi-vessel hydronic network and centralized thermal battery system analyzed in Step 10. The architectures detailed herein establish the deterministic logic loops required by controls engineers and mechatronics integrators to execute transient peak-shaving, dynamic load balancing, and rigid hydronic velocity throttling.

D.A Control Philosophy and Multi-Vessel Operational Modes

To maintain fluid velocity beneath the erosion boundary ($v \leq v_{max}$) while fulfilling transient thermal demands during a batch cycle, the system alternates between four primary operational configurations via automated modulating three-way valves (TCVs) and variable-frequency drive (VFD) circulation loops:

- 1) **Mode I: Baseline Utility Supply (Direct Charging):** Activated when the total process load is below the steady-state boiler capacity ($Q_{load} \leq Q_{steady}$). The thermal battery is structurally isolated, and the utility line directly handles vessel demands using standard feedback proportional-integral-derivative (PID) tuning.
- 2) **Mode II: Peak Shaving / Battery-Assist Discharge:** Prompted automatically when a new process batch initiates and thermal demand spikes past utility capacity ($Q_{load} > Q_{steady}$). The system activates a feed-forward control block, modulating a secondary three-way valve to blend high-temperature fluid from the thermal battery into the loop, maintaining a fixed batch heating rate without overloading the utility.
- 3) **Mode III: Isolated Thermal Battery Recharge:** Triggered during process downtime or low-load intervals. Excess utility energy ($Q_{steady} - Q_{load}$) is diverted into the thermal battery vessel, raising its core internal energy state until upper thermal cutoff boundaries are reached.
- 4) **Mode IV: Multi-Vessel Load Balancing Mode:** Utilizing a distributed cascade architecture, the master Programmable Logic Controller (PLC) monitors real-time temperature differentials ($\Delta T = T_{in} - T_{out}$) across multiple independent vessels. Flow distribution is dynamically scaled using a prioritize-by-deviation algorithm, shifting hydronic volume to vessels exhibiting the steepest temperature drop.

D.B Instrumentation, Sensors, and Actuator Field Profiles

Executing the velocity constraints and energy tracking requirements defined across the core framework requires a synchronized field instrument network. Table XXIX catalogs the sensor characteristics, signal transmission standards, and functional roles mandated within the hydronic loop.

TABLE XXIX
FIELD INSTRUMENTATION AND SIGNAL MAPPING SPECIFICATIONS

Tag ID	Instrument Description	Sensor Type / Element	Signal Protocol	Functional Framework Role
TE-101	Loop Supply Temperature	PT100 RTD (Class A)	4–20 mA / HART	Monitors main utility output profile
TE-102	Battery Core Temperature	Multipoint Thermocouple	4–20 mA / HART	Computes current state-of-charge (SoC)
TE-103	Vessel Batch Temperature	PT100 RTD (Sanitary)	4–20 mA / HART	Feeds the Step 10 transient integration loop
FE-201	Main Line Velocity Sensor	Electromagnetic Flow	Modbus TCP	Validates velocity constraint ($v \leq 2.0$ m/s)
PE-301	Loop Differential Press.	Piezo-Resistive Cell	4–20 mA / HART	Tracks real-time helical head loss (ΔH_f)
TCV-401	Battery Blending Actuator	3-Way Equal Percentage	4–20 mA Loop	Modulates thermal battery discharge volume
VFD-501	Loop Circulation Pump	AC Induction Inverter	Profinet	Alters pump RPM to maintain hydraulic balance

D.C Programmable Logic Controller (PLC) State Transition Matrix

The system logic is implemented within a deterministic state machine to guarantee predictable transitions during sudden multi-vessel load spikes or unexpected system trips. Table XXX provides the logical state matrix used to construct the PLC ladder logic or structured text routines.

TABLE XXX
PLC STATE MACHINE TRANSITION SEQUENCE AND SAFETY INTERLOCKS

Current State	System Profile	Primary Trigger Input	Target Output State	Mandatory Interlock / Safety Guard
01_STANDBY	System idling; pump operating at minimum bypass speed.	Batch initiation command received from HMI terminal.	02_UTILITY_ONLY	Confirm loop pressure PE-301 is within stable static bounds (> 1.5 bar).
02_UTILITY_ONLY	Boiler satisfies loop demand; battery remains isolated.	Process load exceeds utility limit: $Q_{load} > 80$ kW.	03_PEAK_ASSIST	Continuous loop velocity tracking: if $v > 2.0$ m/s, trim VFD-501 immediately.
03_PEAK_ASSIST	TCV-401 modulates open to inject battery energy.	Batch cycle completes; thermal load drops below threshold.	04_RECHARGE	If battery core temperature drops below process target, isolate battery and flag alarm.
04_RECHARGE	Excess utility energy routed into storage matrix.	Battery multipoint sensor TE-102 hits upper ceiling.	01_STANDBY	Disable recharge loop instantly if vessel high-pressure switch trips.
05_FAIL_SAFE	All control valves forced to unpowered safety states.	Fluid velocity limit breach ($v > 2.5$ m/s) or hardware fault.	Safe Shutdown (Lockout)	Bypass all process commands; trigger audible field alarm beacon; dump VFD run signals.

D.D Cascaded Control Loop Proportional-Integral-Derivative (PID) Architecture

To prevent hydraulic hunting when multiple control valves modulate simultaneously, a cascaded control topology is established. The primary master controller monitors the process vessel temperature error ($e(t) = T_{set} - T_{vessel}$) and continuously computes a target hydronic flow rate requirement.

The slave controller receives this flow setpoint and modulates pump speed via VFD-501, referencing electromagnetic flowmeter FE-201 as its direct feedback element. By nesting the velocity loop inside the slower thermal loop, the slave loop can intercept instantaneous pressure fluctuations across the helical network, smoothing out actuator actions and eliminating the risk of accidental erosion velocity overshoots in the system.

D.E HMI / SCADA Screen Layout and Alarm Philosophy

The human-machine interface (HMI) and supervisory control and data acquisition (SCADA) layer serves as the primary operator window into the hydronic network state. Screen organisation follows ISA-101 (HMI Design and Implementation) principles: each display level presents only the information operationally relevant at that level of abstraction, preventing alarm flooding during transient batch events. The hierarchy comprises three display tiers.

Level 1 — System Overview: A single-screen mimic diagram showing all four operational modes (Section D.A), the real-time state of TCV-401 and VFD-501, live readings from TE-101, TE-102, TE-103, FE-201, and PE-301, the thermal battery state-of-charge (SoC) bar expressed as a percentage of $Q_{battery,max}$, and the current PLC state tag (e.g. 03_PEAK_ASSIST). No control actions are executable from this screen; it is display-only to prevent inadvertent mode changes.

Level 2 — Loop Detail Screens: One screen per controlled loop (utility supply, battery discharge, recharge, multi-vessel balancing). Each screen displays the trend history for the relevant temperature, flow, and pressure tags over a configurable 30-minute rolling window, the active setpoint, PV, and output of the governing PID block, and the current interlock status for that loop.

Level 3 — Maintenance and Calibration: Password-protected. Provides manual override capability for individual actuators, PID parameter adjustment, sensor zero/span calibration records, and accumulated run-hour logs for VFD-501.

Table XXXI defines the alarm setpoint philosophy. All alarms are classified per ISA-18.2 (Management of Alarm Systems) into Priority 1 (immediate operator action required), Priority 2 (operator action within the current batch cycle), and Priority 3 (advisory / maintenance). Shelving of Priority 1 alarms is not permitted without a documented bypass authorisation.

TABLE XXXI
ALARM SETPOINT AND PRIORITY CLASSIFICATION MATRIX

Tag	Alarm Description	Lo / LoLo Limit	Hi / HiHi Limit	Priority	Required Operator Action
TE-101	Supply temperature out of band	Lo: $T_{set} - 5^{\circ}\text{C}$ LoLo: $T_{set} - 15^{\circ}\text{C}$	Hi: $T_{set} + 10^{\circ}\text{C}$ HiHi: $T_{set} + 25^{\circ}\text{C}$	P2 / P1	Investigate utility source; P1 triggers Mode IV bypass
TE-102	Battery core temperature deviation	Lo: $T_{bat,min}^{\circ}\text{C}$ LoLo: $T_{bat,min} - 10^{\circ}\text{C}$	Hi: $T_{bat,max}^{\circ}\text{C}$ HiHi: $T_{bat,max} + 5^{\circ}\text{C}$	P2 / P1	P1 forces immediate recharge isolation; flag for inspection
TE-103	Vessel batch temperature deviation	Lo: $T_{proc} - 3^{\circ}\text{C}$	Hi: $T_{proc} + 5^{\circ}\text{C}$ HiHi: $T_{proc} + 12^{\circ}\text{C}$	P3 / P2	P2 triggers setpoint review and valve trim audit
FE-201	Loop flow / velocity breach	Lo: 0.3 m/s (stagnation risk)	Hi: 2.0 m/s HiHi: 2.5 m/s	P2 / P1	P1 trips VFD-501 to minimum; latches 05_FAIL_SAFE
PE-301	Differential pressure loss of loop	LoLo: 0.5 bar (possible leak)	HiHi: 6.0 bar (blockage)	P1	Isolate loop; inspect coil for fouling or mechanical failure
VFD-501	Drive fault or overtemperature	—	Any drive fault code	P1	Acknowledge; switch to manual bypass pump if installed
TCV-401	Valve position feedback mismatch	$ \text{Commanded} - \text{Actual} > 5\%$ for > 10 s		P2	Check pneumatic/electric actuator supply and positioner

Deadband of $\pm 0.5^{\circ}\text{C}$ applied to all temperature alarms to suppress chattering during stable steady-state operation. HiHi and LoLo thresholds trigger automatic PLC state transitions; Hi and Lo thresholds annunciate only.

D.F PID Tuning Parameters and Recommended Gain Values

The cascaded control topology described in Section D.D requires independently tuned PID blocks for each loop layer. Because the thermal time constants of a large hydronic vessel are significantly slower than the hydraulic time constants of the circulation loop, the two loops operate on different update rates and must not share tuning parameters. All PID blocks are implemented in the positional (non-velocity) form with output clamping and anti-windup via integrator back-calculation, as given by:

$$u(t) = K_p \left[e(t) + \frac{1}{T_i} \int_0^t e(\tau) d\tau + T_d \frac{de(t)}{dt} \right] \quad (107)$$

where K_p is the proportional gain, T_i the integral time constant (seconds), T_d the derivative time constant (seconds), and $e(t) = SP - PV$ the instantaneous error. Derivative action is applied to the process variable only (not the error) to prevent derivative kick on setpoint steps. Table XXXII provides the recommended starting gain values derived from open-loop step-test characterisation of a representative single-vessel hydronic system. Site-specific commissioning using the Cohen-Coon or IMC tuning method is required to confirm final values before live operation.

TABLE XXXII
RECOMMENDED PID TUNING PARAMETERS BY CONTROL LOOP

Loop Name	Controlled Variable	Actuator	K_p	T_i (s)	T_d (s)	Notes
Master Thermal (Outer)	Vessel batch temperature TE-103	TCV-401 position setpoint	1.8–2.5	180–300	15–30	Slow integrating process; low K_p prevents overshoot on batch start
Slave Hydraulic (Inner)	Loop flow velocity FE-201	VFD-501 speed (%)	0.8–1.4	8–20	1–3	Fast loop; T_d active to suppress pressure transient spikes
Battery Discharge	Battery blend ratio via TCV-401	TCV-401 direct	2.0–3.0	60–120	0	Feed-forward dominant; integral only for steady-state trim; no derivative
Recharge Loop	Battery core temperature TE-102	Recharge divert valve	1.2–1.8	240–480	0	Very slow thermal mass; derivative omitted to avoid noise amplification
Multi-Vessel Balancing	ΔT deviation per vessel	Per-vessel VFD trim	0.5–1.0	30–60	0	Proportional-integral only; gains intentionally low to prevent inter-vessel hunting

Output limits: Master loop 0–100% valve position; Slave loop 20–100% VFD speed (20% minimum to prevent coil stagnation). Anti-windup back-calculation time constant: $T_{aw} = \sqrt{T_i \cdot T_d}$ for loops with derivative; $T_{aw} = 0.1 \cdot T_i$ otherwise.

The field instrument and actuator network is organised as a three-layer communication hierarchy. Physical isolation between layers is mandatory to prevent process-level noise from propagating into the supervisory layer and to maintain deterministic scan times on the PLC execution bus.

Layer 1 — Field Bus (Process Level): All analogue field instruments (TE-101, TE-102, TE-103, PE-301, TCV-401) communicate over the 4–20 mA analogue loop with superimposed HART (Highway Addressable Remote Transducer) digital overlay. HART communication is used exclusively for remote configuration, diagnostics, and secondary variable transmission; the primary process variable is always carried by the analogue signal to guarantee scan-cycle independence. Shielded twisted-pair (STP) cable with drain wire grounded at the marshalling cabinet end only is mandatory for all HART runs to prevent ground loops.

Layer 2 — Control Bus (PLC Level): The electromagnetic flowmeter FE-201 and the variable-frequency drive VFD-501 communicate over Profinet IO (IEC 61158 / IEC 61784), providing cycle times of 1–4 ms suitable for the inner hydraulic control loop. The master PLC also hosts a Modbus TCP server on a dedicated Ethernet port for inter-PLC data exchange in multi-vessel balancing configurations. Each vessel PLC is a Modbus TCP slave, publishing its live tag values (temperature, flow, SoC) to the master at a 500 ms poll interval.

Layer 3 — Supervisory Bus (SCADA Level): The HMI/SCADA server communicates with the master PLC via OPC-UA (IEC 62541) over a dedicated VLAN, physically separated from the control bus. The OPC-UA server exposes a read/write node space for all operator-accessible setpoints and a read-only node space for historian data logging. SCADA historian sampling is configured at 1-second resolution for all analogue tags and event-driven for all discrete state changes.

Table XXXIII summarises the protocol assignments, physical media, and performance specifications for each network layer.

TABLE XXXIII
COMMUNICATION PROTOCOL AND NETWORK LAYER SPECIFICATIONS

Layer	Protocol	Physical Medium	Topology	Cycle / Poll Rate	Connected Devices
Layer 1	4–20 mA + HART (rev 7)	STP 2-core, 0.5 mm ²	Point-to-point star to marshalling	N/A (analogue continuous)	TE-101/102/103, PE-301, TCV-401
Layer 2	Profinet IO (RT Class 1)	Cat6A STP Ethernet	Line / star hybrid	1–4 ms	FE-201, VFD-501
Layer 2	Modbus TCP (slave nodes)	Cat6A STP Ethernet, isolated switch	Star from master PLC	500 ms poll	Satellite vessel PLCs (Mode IV)
Layer 3	OPC-UA (IEC 62541)	Cat6A, dedicated VLAN	Star from SCADA server	1 s historian; event-driven alarms	HMI terminals, SCADA historian, engineering workstation

All Ethernet segments must be separated by managed switches with VLAN tagging; unmanaged switches are not permitted on Layer 2.
Cybersecurity hardening per IEC 62443-3-3 Security Level 2 is recommended for any OPC-UA endpoint exposed beyond the local control network.

D.H Failure Mode and Effects Analysis (FMEA) and Fault Response

A structured failure mode and effects analysis is required to validate that the PLC state machine defined in Section D.C captures all credible fault conditions and that no single-point failure can result in uncontrolled erosion velocity, unintended thermal battery discharge, or unsafe vessel pressurisation. The FMEA is structured per IEC 60812 (Analysis Techniques for System Reliability). Each failure mode is assigned a Risk Priority Number (RPN) computed as the product of Severity (S), Occurrence likelihood (O), and Detectability (D), each scored on a 1–10 scale. Modes with $RPN \geq 100$ are classified as critical and require a dedicated hardware interlock independent of the PLC software layer. Table XXXIV documents the primary failure modes identified for the hydronic control system.

TABLE XXXIV
FAILURE MODE AND EFFECTS ANALYSIS (FMEA) — HYDRONIC CONTROL SYSTEM

Failure Mode	Root Cause	Effect on System	S	O	D	RPN	Recommended Mitigation
FE-201 signal loss	Cable fault or transmitter failure	Inner PID loop loses flow feedback; VFD runs open-loop	8	3	4	96	Watchdog timer on FE-201 signal; fallback to minimum VFD speed on loss of signal
TCV-401 stuck open	Actuator spring or positioner failure	Uncontrolled battery discharge; vessel overtemperature	9	2	3	54	Mechanical position indicator; TE-103 HiHi trips battery isolation block valve
TCV-401 stuck closed	Positioner or pneumatic supply failure	Battery cannot discharge; thermal demand unmet during peak	5	2	3	30	TE-103 Lo alarm prompts operator; manual bypass line with block valve
VFD-501 trip / fault	Overtemperature, overcurrent, or supply fault	Immediate loss of loop circulation; coil stagnation risk	9	3	2	54	VFD fault output hardwired to PLC DI; auto-transition to 05_FAIL_SAFE; standby pump relay if installed
Loop over-velocity event	VFD runaway or control loop instability	Erosion of coil inner wall; long-term hydraulic degradation	10	2	2	40	Hardware overspeed relay on FE-201 independent of PLC; trips VFD contactor directly
TE-102 multipoint failure	Thermocouple open-circuit or moisture ingress	SoC calculation invalid; risk of battery over-charge	7	4	4	112	Critical ($RPN \geq 100$): Redundant thermocouple pair on battery core; majority-vote logic in PLC
PLC processor fault	Hardware fault or watchdog timeout	All automated control outputs frozen at last state	10	1	2	20	Hot-standby PLC in duplex configuration; all control valves fail-safe to defined de-energised positions
Modbus TCP link loss (Mode IV)	Network switch fault or cable break	Master loses satellite vessel data; balancing algorithm degrades	6	3	3	54	Satellite PLCs revert to local standalone PID on comms timeout (>3 consecutive missed polls)
Loop pressure loss (PE-301 LoLo)	Coil or pipe leak	Risk of fluid escape and thermal hazard	9	2	2	36	PE-301 LoLo hardwired to master isolation valve; 05_FAIL_SAFE latch; field alarm beacon
SCADA OPC-UA server loss	Server crash or network VLAN fault	Loss of operator visibility; control continues autonomously	4	3	2	24	PLC operates independently of SCADA; local HMI panel on PLC maintains Level 1 display

Severity, Occurrence, and Detectability scored 1 (lowest) to 10 (highest) per IEC 60812. $RPN = S \times O \times D$.

All $RPN \geq 100$ items require hardware interlock independent of PLC software layer and must be re-assessed after any system modification.

APPENDIX E

PLC IMPLEMENTATION AND DISCRETE-TIME CONTROL FORMULATIONS

This appendix translates the continuous-time thermal logic established in the sizing framework into the discrete-time difference equations required by industrial microprocessors. The formulations provided herein are specifically formatted for Programmable Logic Controllers (PLCs) and Distributed Control Systems (DCS), ensuring safe actuator modulation, electrical noise rejection, and system stability under massive thermal inertia.

E.A Discrete-Time Proportional-Integral-Derivative (PID) Algorithms

The theoretical continuous-time PID algorithm is defined across the complex domain as:

$$u(t) = K_p e(t) + K_i \int_0^t e(\tau) d\tau + K_d \frac{de(t)}{dt}$$

Because a PLC samples data at fixed, discrete scan intervals (Δt), this continuous integral must be converted algebraically using backward Euler approximations. For hydronic valves and Variable Frequency Drives (VFDs) which require bumpless transfers and smooth actuation, the **Velocity (Incremental) PID Form** is utilized. Instead of calculating the absolute output $u(k)$, the PLC calculates the incremental change required for the current scan cycle $\Delta u(k)$:

$$\Delta u(k) = K_p [e(k) - e(k-1)] + K_i \Delta t e(k) + \frac{K_d}{\Delta t} [e(k) - 2e(k-1) + e(k-2)]$$

The final actuator signal sent to the field hardware is then accumulated:

$$u(k) = u(k-1) + \Delta u(k)$$

E.B Derivative Kick Mitigation and Integral Anti-Windup Clamping

Derivative Kick Avoidance: If a plant operator suddenly alters the process setpoint (SP) via the Human Machine Interface (HMI), the error $e(k)$ shifts instantaneously. This triggers an infinite spike in the standard derivative term, violently forcing the VFD to maximum frequency. To prevent this, the algorithm is modified to take the derivative of the Process Variable (PV) instead of the Error (e):

$$\text{Derivative Term} = -\frac{K_d}{\Delta t} [PV(k) - 2PV(k-1) + PV(k-2)]$$

Integral Anti-Windup (Conditional Clamping): Large thermal batch processes possess massive inertia, causing the integral term to accumulate indefinitely while waiting for the fluid to heat up. To prevent the loop from “winding up” and overshooting the target, the PLC logic actively freezes the *incremental* integral contribution $\Delta I(k)$ for the current scan cycle whenever the actuator output has reached its physical saturation boundary. The accumulated output $u(k)$ is not zeroed; only the increment for that cycle is suppressed:

$$\text{If } u(k) \geq u_{max} \text{ and } e(k) > 0: \quad \Delta I(k) = 0$$

$$\text{If } u(k) \leq u_{min} \text{ and } e(k) < 0: \quad \Delta I(k) = 0$$

This formulation is consistent with the velocity-form ST implementation in Section E.E, where I_Term represents the per-scan integral increment $K_i \cdot \Delta t \cdot e(k)$, not the total accumulated sum.

E.C First-Order Low-Pass Sensor Filtering (EMA)

Industrial fluid environments induce turbulent noise across RTDs (TE-101) and electromagnetic flowmeters (FE-201). Before entering the PID calculation block, raw analog signals are smoothed using a discrete Exponential Moving Average (EMA) low-pass filter:

$$PV_{filtered}(k) = \alpha \cdot PV_{raw}(k) + (1 - \alpha) \cdot PV_{filtered}(k-1)$$

Where the smoothing constant α is derived from the scan rate (Δt) and the desired filter time constant (τ_f):

$$\alpha = \frac{\Delta t}{\tau_f + \Delta t} \quad (0 < \alpha \leq 1)$$

E.D PLC Execution Interval and Hardware Scan Strategy

Executing complex floating-point PID math on every millisecond PLC hardware cycle needlessly consumes CPU bandwidth and amplifies derivative noise. The control task execution interval (Δt_{scan}) is determined by the thermal time constant of the battery vessel (S_t), computed from the physical system parameters in Step 10. Based on the Nyquist–Shannon constraint applied to the dominant thermal inertia, the recommended logic block execution interval is:

$$\Delta t_{scan} \approx \frac{S_t}{20}$$

As a representative reference: for a system where Step 10 yields $S_t = 6665$ s, this gives $\Delta t_{scan} \approx 330$ s (5.5 minutes). The divisor of 20 provides a minimum of ten sample points per half-period of the slowest thermal oscillation mode, consistent with standard process control practice. For systems with significantly smaller thermal mass, $S_t/20$ must be re-evaluated and the scan interval reduced accordingly.

E.E IEC 61131-3 Structured Text (ST) Implementation — PID Control Block

The following standardized ST logic encapsulates the discrete algorithms established above, ready for implementation in Siemens TIA Portal, Beckhoff TwinCAT, or Rockwell Studio 5000 environments.

```
1  VAR
2      // Inputs and Parameters
3      PV_Raw      : REAL; // Raw Analog Sensor Input
4      SP          : REAL; // Target Setpoint
5      Kp, Ki, Kd   : REAL; // Tuning Parameters
6      dt          : REAL; // Scan time in seconds (= St / 20, from Step 10)
7      Alpha       : REAL; // Filter Constant (= dt / (tau_f + dt))
8
9      // Memory States (RETAIN: preserved across PLC power cycle)
10     PV_Filtered  : REAL;
11     PV_Prev1    : REAL;
12     PV_Prev2    : REAL;
13     Error_Prev  : REAL;
14     Actuator_Out: REAL;
15
16     // Intermediate Calculation Variables
17     Error        : REAL;
18     Delta_U      : REAL;
19     P_Term       : REAL;
20     I_Term       : REAL;
21     D_Term       : REAL;
22 END_VAR
23
24 // --- 1. Exponential Moving Average Sensor Filter ---
25 PV_Filtered := (Alpha * PV_Raw) + ((1.0 - Alpha) * PV_Filtered);
26
27 // --- 2. Error Calculation ---
28 Error := SP - PV_Filtered;
29
30 // --- 3. Proportional Term (error-based) ---
31 P_Term := Kp * (Error - Error_Prev);
32
33 // --- 4. Derivative Term (PV-based to prevent setpoint-step kick) ---
34 D_Term := -(Kd / dt) * (PV_Filtered - (2.0 * PV_Prev1) + PV_Prev2);
35
36 // --- 5. Integral Increment with Anti-Windup Clamping ---
37 // Suppresses only the per-scan increment; does not zero the accumulator.
38 IF (Actuator_Out >= 100.0 AND Error > 0.0) OR
39    (Actuator_Out <= 0.0 AND Error < 0.0) THEN
40     I_Term := 0.0; // Freeze increment this scan; accumulated u(k) unchanged
41 ELSE
42     I_Term := Ki * dt * Error;
43 END_IF;
44
45 // --- 6. Velocity Form Output Accumulation ---
46 Delta_U := P_Term + I_Term + D_Term;
47 Actuator_Out := Actuator_Out + Delta_U;
48
49 // --- 7. Absolute Output Saturation Limits ---
50 IF Actuator_Out > 100.0 THEN Actuator_Out := 100.0; END_IF;
51 IF Actuator_Out < 0.0 THEN Actuator_Out := 0.0; END_IF;
52
53 // --- 8. Update Memory Registers for Next Scan ---
54 Error_Prev := Error;
55 PV_Prev2 := PV_Prev1;
56 PV_Prev1 := PV_Filtered;
```

Listing 1. Velocity Form PID with Anti-Windup and EMA Filtering

E.F Ladder Logic (LD) Implementation Guidelines

While Structured Text (ST) natively supports complex algebraic difference equations, many legacy or visually-oriented plant environments require IEC 61131-3 Ladder Diagram (LD) architecture. When implementing the VCH control loops in LD, the following structural guidelines apply:

- **Cyclic Execution via Interrupts:** Instead of using standard internal timers (TON) to trigger the math, the LD network should reside inside a dedicated cyclic hardware interrupt block (e.g., OB30 in Siemens TIA Portal or a Periodic Task in Rockwell RSLogix). The block execution rate must be statically configured to the Δt_{scan} interval calculated in E.D.
- **Mathematical Compute Blocks:** The EMA sensor filter cannot easily be executed with basic add/multiply coils. A dedicated Compute (CPT) instruction block should be utilized to calculate the $(1 - \alpha)$ filtering formula in a single rung.
- **Native PID Instruction Utilization:** Rather than writing the full velocity-form equation manually in LD, utilize the native PID instruction blocks (such as PID_Compact or PIDE). Ensure the block parameters are explicitly configured to read the *Filtered* PV rather than the *Raw* physical analog channel.

- **Boolean Anti-Windup Triggers:** Use standard contact structures (XIC/XIO) to evaluate actuator saturation. If the PID instruction block output reaches 100% and the error remains positive, route a parallel rung branch to a MOVE (MOV) instruction that immediately forces the integral sum register to zero.

E.G Hardwired Relay-Based Control (Non-PLC Architecture)

For smaller, isolated hydronic skids, or as a robust fail-safe override to the primary PLC, a traditional hardwired electromechanical control architecture can be employed without microprocessor intervention.

- **Bang-Bang (On/Off) Control Profiles:** Proportional-Integral modulation is impossible without a dedicated controller. Instead, continuous analog temperature sensors (RTDs) are replaced with mechanical thermostats or liquid-filled capillary thermal switches. When the bulk fluid temperature drops below the physical setpoint, the mechanical switch snaps closed.
- **Relay Logic and Contactors:** The closure of the thermostat switch energizes a 24 VDC or 120 VAC Master Control Relay (MCR) coil. The MCR subsequently pulls in the heavy-duty main motor contactor, providing three-phase power directly to the primary utility heater or the fixed-speed circulation pump.
- **Hardware Safety Interlocks:** To mimic the PLC's software velocity and flow safeguards, physical flow switches (paddle or thermal dispersion type) and high-limit pressure switches (capsule type) are wired directly in series with the MCR coil. If flow drops unexpectedly or system pressure surges beyond mechanical limits, the series circuit breaks instantly, dropping the contactor and halting the hydronic loop entirely.
- **Operational Limitations:** While hardwired relays are highly robust and immune to software faults, network crashes, and EMI corruption, they suffer from severe thermal hysteresis (temperature swinging). Furthermore, relay logic cannot execute dynamic load-balancing or thermal battery peak shaving (Modes II and IV). A hardwired network restricts the hydronic framework exclusively to baseline direct charging and standby states.

E.H Watchdog Timer and 05_FAIL_SAFE State Implementation

The PLC state machine defined in Appendix D (Section D.C) designates 05_FAIL_SAFE as the terminal safety state, reachable from any operating mode upon detection of a velocity limit breach, hardware fault, or loss of a critical field signal. Reaching this state via software logic alone is insufficient for a safety-critical hydronic loop: a PLC processor fault, scan-cycle overrun, or memory corruption event could prevent the state transition from executing entirely. Two complementary mechanisms are therefore required.

Software Watchdog (PLC Internal): On every scan cycle in which the control block executes normally, the ST block writes a heartbeat token to a dedicated RETAIN boolean flag (WD_Heartbeat). A separate watchdog timer task, scheduled at a faster interrupt priority than the main control task, monitors this flag. If the heartbeat is not refreshed within two consecutive watchdog periods, the task unconditionally forces all control outputs to their de-energised fail-safe positions and latches the FAIL_SAFE_Active flag, which is hardwired to the field alarm beacon and the VFD run-enable relay.

Hardware Watchdog (Independent of PLC): The VFD run-enable signal (VFD-501 enable input) is routed through an external hardware watchdog relay module, independent of the PLC output card. This relay requires a continuous 1 Hz pulse train from the PLC digital output to remain energised. If the pulse train stops for any reason — including PLC processor halt, output card failure, or loss of 24 VDC control supply — the relay drops out, removing the run-enable signal from the drive and bringing the loop to a controlled deceleration stop.

The ST implementation below covers the software watchdog block and the 05_FAIL_SAFE state transition logic, intended to execute in a high-priority cyclic interrupt task at a fixed 1-second interval, independent of the main 5.5-minute control task.

```

1  VAR
2      // Watchdog Coordination (set by main PID block each scan)
3      WD_Heartbeat      : BOOL;          // Written TRUE by PID block each execution
4      WD_Prev           : BOOL;          // Previous-scan heartbeat state
5      WD_MissCount      : UINT := 0;     // Consecutive missed heartbeat counter
6      WD_TripThreshold  : UINT := 2;     // Trips FAIL_SAFE after N missed scans
7
8      // Velocity Monitoring (read from FE-201 signal chain)
9      Velocity_Live     : REAL;          // Current loop velocity in m/s
10     Velocity_HiHi     : REAL := 2.5;   // Hard erosion ceiling (Appendix D alarm)
11
12     // VFD Hardware Pulse Output (1 Hz square wave to external WD relay)
13     HW_WD_Pulse       : BOOL;          // Physical DO to hardware watchdog relay
14     HW_WD_Toggle      : BOOL;          // Internal toggle state
15
16     // System State Flags
17     FAIL_SAFE_Active   : BOOL := FALSE; // Latching; requires manual HMI reset
18     PLC_State          : UINT;          // Mirrors D.C state machine (05 = FAIL_SAFE)
19
20     // Safe-State Actuator Demand Outputs
21     TCV401_Demand     : REAL;          // 0.0 = fully closed (battery isolated)
22     VFD501_Demand     : REAL;          // 0.0 = stop command to drive
23     Alarm_Beacon      : BOOL;          // Hardwired field alarm output
24 END_VAR
25
26 // -----
27 // BLOCK 1: Hardware Watchdog Pulse Generator (1 Hz toggle)

```

```

28 // The external relay module requires continuous alternation to stay closed.
29 // If this block stops executing, the relay drops and the VFD loses run-enable.
30 // -----
31 HW_WD_Toggle := NOT HW_WD_Toggle;
32 HW_WD_Pulse := HW_WD_Toggle; // Written to physical DO each second
33
34 // -----
35 // BLOCK 2: Software Heartbeat Monitor
36 // Checks whether the main PID task refreshed WD_Heartbeat since last scan.
37 // -----
38 IF WD_Heartbeat = WD_Prev THEN
39     // Heartbeat has not changed: main task missed its execution window
40     WD_MissCount := WD_MissCount + 1;
41 ELSE
42     // Heartbeat toggled normally: reset miss counter
43     WD_MissCount := 0;
44 END_IF;
45 WD_Prev := WD_Heartbeat; // Capture state for next comparison
46
47 // -----
48 // BLOCK 3: FAIL_SAFE Trigger Evaluation
49 // Conditions that unconditionally latch 05_FAIL_SAFE:
50 // (a) Watchdog miss count exceeded threshold (main task unresponsive)
51 // (b) Live loop velocity breaches the HiHi erosion ceiling
52 // -----
53 IF (WD_MissCount >= WD_TripThreshold) OR
54 (Velocity_Live > Velocity_HiHi) THEN
55     FAIL_SAFE_Active := TRUE; // Latching: only cleared by operator HMI reset
56 END_IF;
57
58 // -----
59 // BLOCK 4: Safe-State Output Enforcement
60 // When latched, all actuator demands are forced to de-energised positions
61 // regardless of any other running logic block or PID output.
62 // -----
63 IF FAIL_SAFE_Active THEN
64     PLC_State := 5; // Transition to 05_FAIL_SAFE per D.C matrix
65     TCV401_Demand := 0.0; // TCV-401 forced closed: battery fully isolated
66     VFD501_Demand := 0.0; // VFD-501 run demand removed: loop decelerates
67     Alarm_Beacon := TRUE; // Field audible/visual alarm energised
68     HW_WD_Pulse := FALSE; // Pulse train halted: hardware relay drops out
69     // independently removing VFD run-enable
70 END_IF;

```

Listing 2. Software Watchdog and 05_FAIL_SAFE State Transition Logic

Two implementation requirements must be confirmed during factory acceptance testing (FAT) before commissioning. First, the hardware watchdog relay drop-out time must be verified to be shorter than the VFD deceleration ramp time, ensuring the drive reaches zero speed before any mechanical brake or isolation valve actuates. Second, the FAIL_SAFE_Active latch must be confirmed as non-resettable from any automated logic path; reset must require an explicit operator acknowledgement action on the Level 3 HMI screen defined in Appendix D, Section D.E.